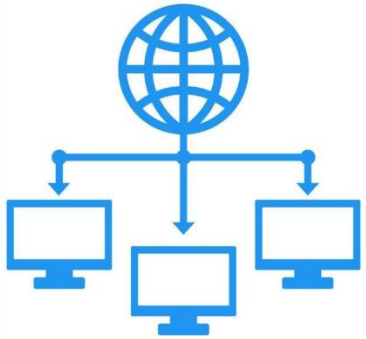


Networking in Cybersecurity 101

Basics of Computer Network
Networking Knowledge
Common Protocols and Their usage
Common Ports and Their usage
SSL and TLS Basics
Public IP vs Private IP
Network Topologies

- Star
- Ring
- Mesh
- Bus

Understanding Common Protocols

- Ssh
- RDP
- FTP
- SFTP
- HTTP/HTTPS

Network tools

- nslookup
- iptables
- ifconfig/ipconfig
- ping
- netstat
- tracert/traceroute

Abbas Aghayev

2026

Basics of Computer Networking

Computer networking refers to the practice of connecting two or more computing devices, creating an infrastructure in which they can exchange data, resources, and software. It is a fundamental part of cyber security and IT skills. In this chapter, we will cover five aspects of computer networking, including networking devices, network types, network protocols, IP addresses, and the OSI model.

Networking Devices

Several devices enable and facilitate communication between different devices. Common networking devices include:

- **Hubs**: Devices that connect different devices together, transmitting data packets to all devices on the network.
- **Switches**: Similar to hubs, but transmit data packets only to specific devices instead of broadcasting to all.
- **Routers**: Devices that direct data packets between networks and provide the best path for data packets to reach their destination.
- **Firewalls**: Devices or software that monitor and filter incoming and outgoing network traffic, allowing only authorized data to pass through.

Network Types

There are various types of networks based on the distance they cover, and the number of devices they connect. A few common network types are:

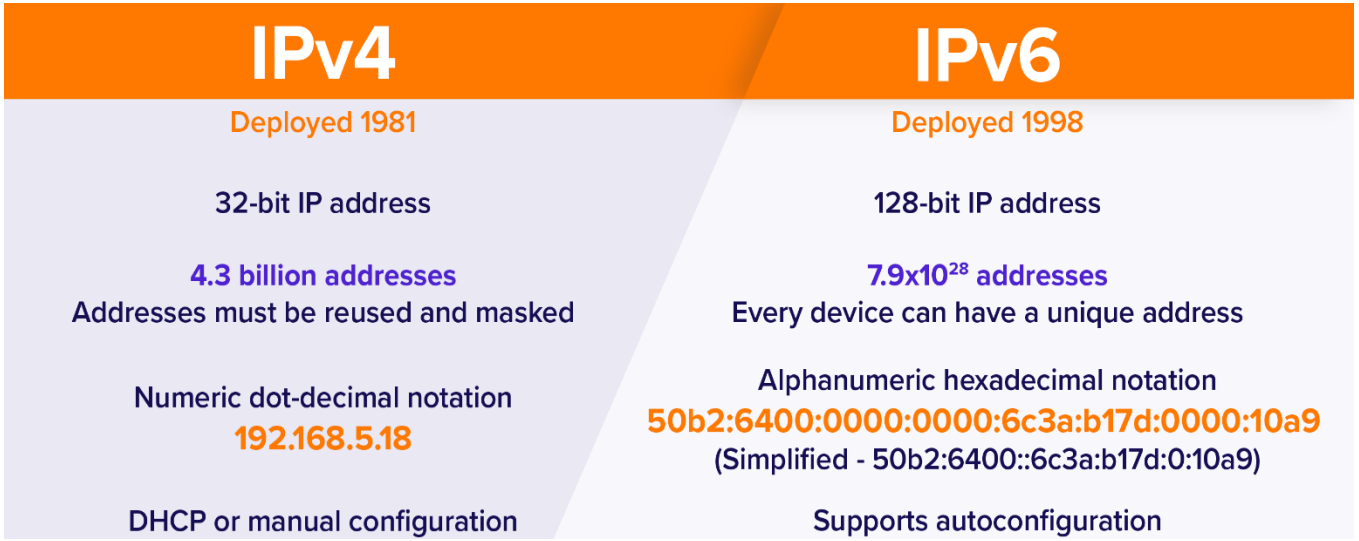
- **Personal Area Network (PAN)**: Connects devices within an individual workspace, typically within a range of 10 meters.
- **Local Area Network (LAN)**: Covers a small geographical area, such as a home or office, connecting multiple computers and other devices.
- **Wide Area Network (WAN)**: Covers a larger geographical area, interconnecting different LANs, often using leased telecommunication lines or wireless links.
- **Virtual Private Network (VPN)**: A secure network established over the public internet, encrypting the data transferred and restricting access to authorized users only.

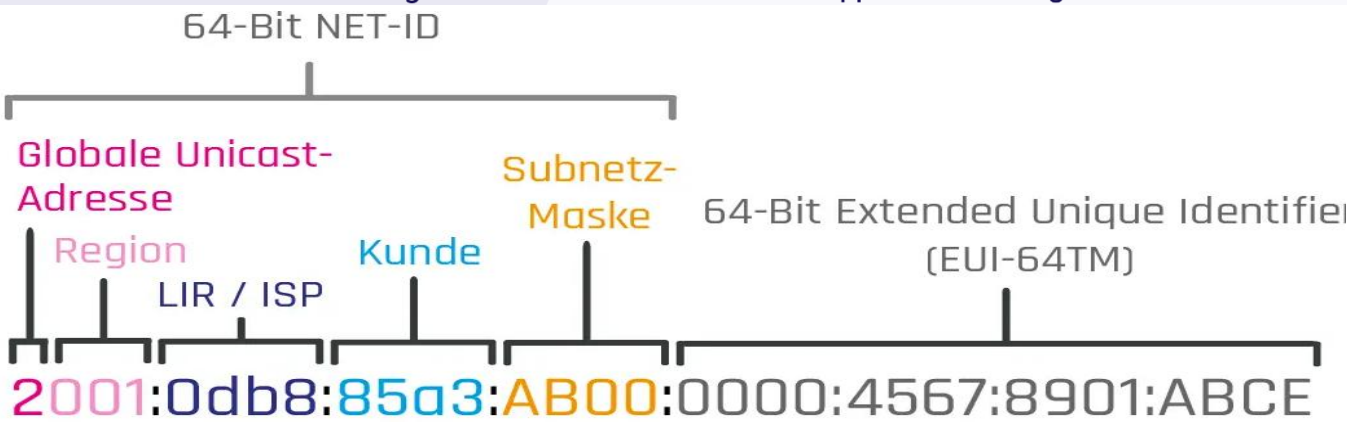
Network Protocols

Protocols are sets of rules that govern the communication between devices within a network. Some of the most common protocols include:

- **Internet Protocol (IP): Facilitates the transmission of data packets, assigning unique IP addresses to identify devices.**
- **Transmission Control Protocol (TCP)**: Ensures the reliable transmission of data and establishes connections between devices.
- **User Datagram Protocol (UDP)**: A lightweight, fast, but less reliable protocol compared to TCP, often used for streaming and gaming applications.

IP Addresses





An Internet Protocol address is a unique identifier assigned to every device in a network. There are two types (versions) of IP addresses:

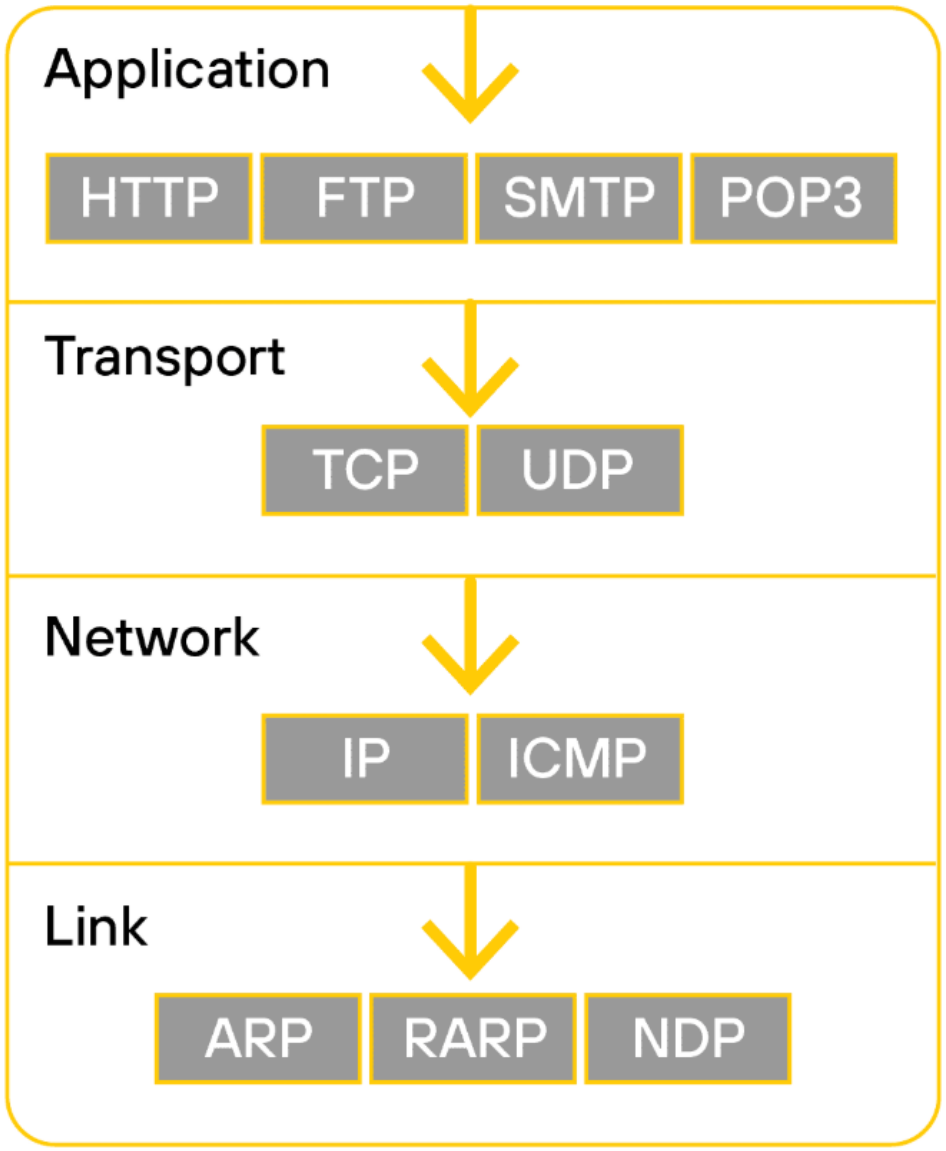
- **IPv4**: Uses a 32-bit addressing system, allowing approximately 4.3 billion unique IP addresses.
- **IPv6**: Uses a 128-bit addressing system, providing a significantly larger number of available IP addresses.

IP addresses can also be categorized as dynamic or static, depending on whether they change over time or remain constant for a device.

TCP/IP Protocol Stack

The TCP stack places the block of data into an output buffer on the server and determines the maximum segment size of individual TCP blocks permitted by the server operating system. The TCP stack then divides the data blocks into appropriately sized segments, adds a TCP header, and sends the segment to the IP stack on the server.

The IP stack adds source and destination IP addresses to the TCP segment and notifies the server operating system that it has an outgoing message that is ready to be sent across the network. When the server operating system is ready, the IP packet is sent to the network adapter, which converts the IP packet to bits and sends the message across the network.



IPv4 Structure

The 32-bit address space (four octets) of an IPv4 address limits the total number of unique public IP addresses to about 4.3 billion. In 2018, the pool of available IPv4 addresses that can be assigned to organizations was officially depleted. A small pool of IPv4 addresses was reserved by each regional internet registry to facilitate the transition to IPv6.

Dotted Decimal Notation
192.168.12.24

Binary Notation
11000000.10101000.0000
1100.00011000

An example of an IPv4 address in dotted decimal notation and binary notation.

Numbering System

Before following the path data takes across internetworks, you should understand how network systems are addressed. Physical, logical, and virtual addressing in computer networks require a basic understanding of decimal (base 10), hexadecimal (base 16), and binary (base 2) numbering.

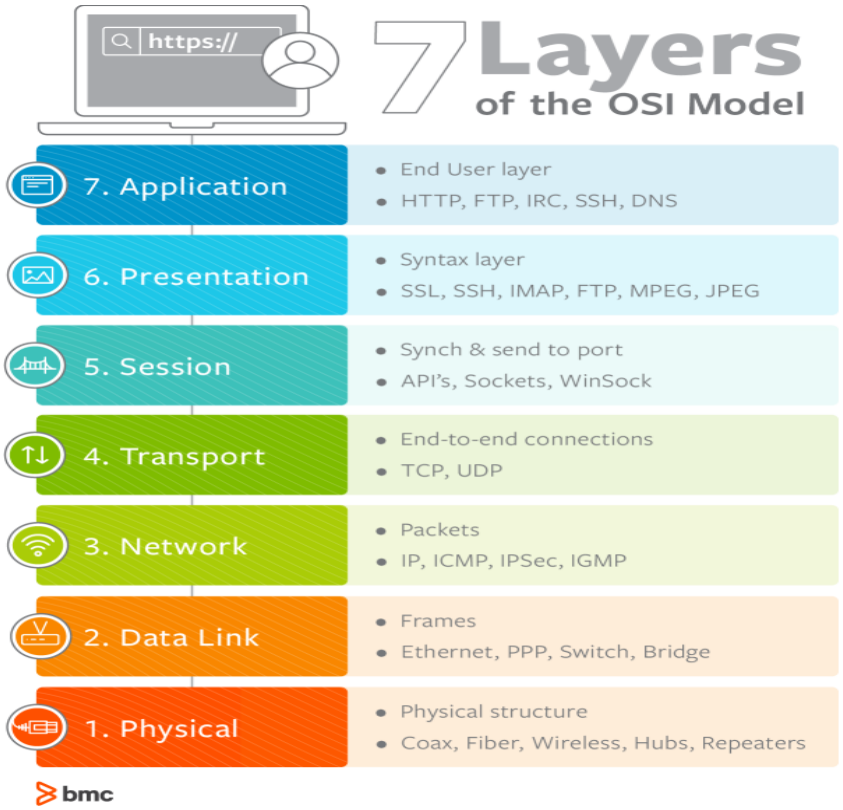
Decimal	Hexadecimal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

OSI Model

The Open Systems Interconnection (OSI) model is a conceptual framework used to understand and describe how different network protocols interact. It divides networking functions into seven distinct layers:

- **Physical Layer**: Deals with the physical connection between devices, including cabling and hardware.
- **Data Link Layer**: Handles the communication between adjacent devices on the same network.
- **Network Layer**: Identifies the best route for data packets and manages IP addresses.
- **Transport Layer**: Ensures the reliable transmission of data, including error checking and flow control.
- **Session Layer**: Establishes, maintains, and terminates connections between applications on different devices.
- **Presentation Layer**: Translates data into a format that is suitable for transmission between devices.
- **Application Layer**: Represents the user interface with which applications interact.

Mastering the basics of computer networking is key to understanding and implementing effective cyber security measures. This chapter has covered essential networking concepts, but it is important to continually expand your knowledge in this ever-evolving field.



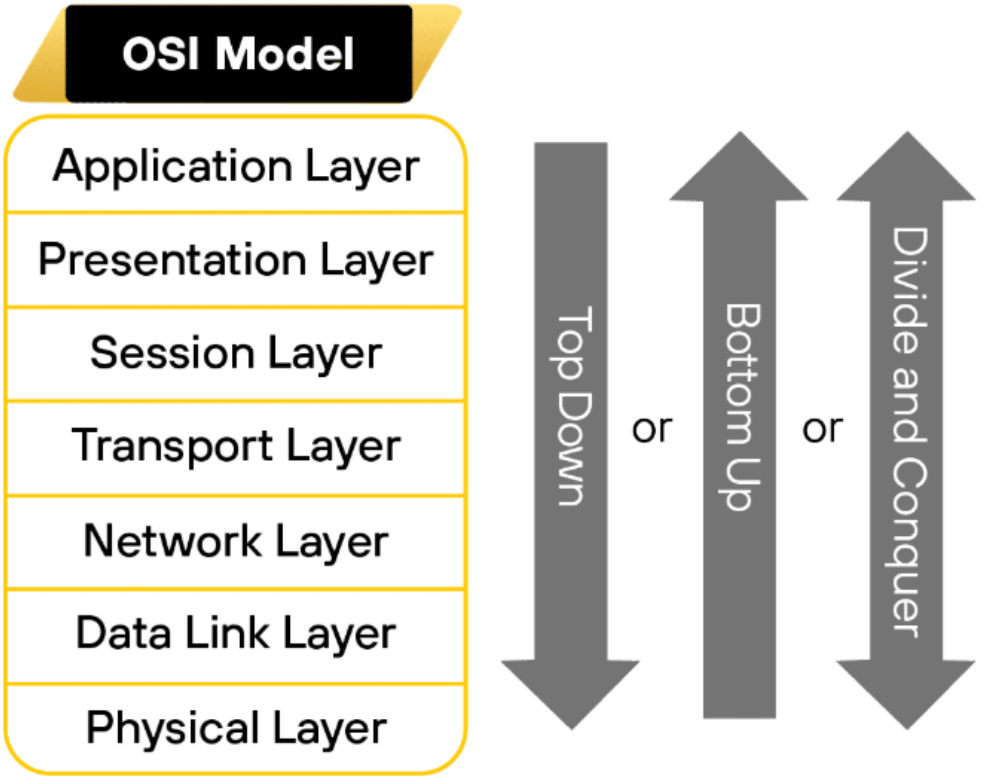
OSI (Open Source Interconnection) 7 Layer Model

Layer	Application/Example		Central Device/ Protocols		DOD4 Model
Application (7) Serves as the window for users and application processes to access the network services.	**End User layer** Program that opens what was sent or creates what is to be sent Resource sharing • Remote file access • Remote printer access • Directory services • Network management		**User Applications** SMTP	GATEWAY Can be used on all layers	Process
Presentation (6) Formats the data to be presented to the Application layer. It can be viewed as the "Translator" for the network.	**Syntax layer** encrypt & decrypt (if needed) Character code translation • Data conversion • Data compression • Data encryption • **Character Set Translation**		JPEG/ASCII EBDIC/TIFF/GIF PICT		
Session (5) Allows session establishment between processes running on different stations.	**Synch & send to ports** (logical ports) Session establishment, maintenance and termination • Session support - perform security, name recognition, logging, etc.		**Logical Ports** RPC/SQL/NFS NetBIOS names		
Transport (4) Ensures that messages are delivered error-free, in sequence, and with no losses or duplications.	**TCP** Host to Host, Flow Control Message segmentation • Message acknowledgement • Message traffic control • Session multiplexing	PACKET FILTERING	TCP/SPX/UDP		Host to Host
Network (3) Controls the operations of the subnet, deciding which physical path the data takes.	**Packets** ("letter", contains IP address) Routing • Subnet traffic control • Frame fragmentation • Logical-physical address mapping • Subnet usage accounting		**Routers** IP/IPX/ICMP		Internet
Data Link (2) Provides error-free transfer of data frames from one node to another over the Physical layer.	**Frames** ("envelopes", contains MAC address) [NIC card — Switch — NIC card] (end to end) Establishes & terminates the logical link between nodes • Frame traffic control • Frame sequencing • Frame acknowledgment • Frame delimiting • Frame error checking • Media access control		**Switch Bridge WAP** PPP/SLIP	Land Based Layers	Network
Physical (1) Concerned with the transmission and reception of the unstructured raw bit stream over the physical medium.	**Physical structure** Cables, hubs, etc. Data Encoding • Physical medium attachment • Transmission technique - Baseband or Broadband • Physical medium transmission Bits & Volts		**Hub**		

OSI Model: Explaining Network Activity in a Logical Structure

The OSI model provides a logical model for troubleshooting complex host and network issues. Depending on the situation, you might use the bottom-up, top-down, or divide-and-conquer approach when you use the OSI model to guide your troubleshooting efforts.

In other situations, you might make an educated guess about the source of the issue and begin investigating at the corresponding layer of the OSI model. You could also use the substitution method (replacing a bad component with a known good component) to quickly identify and isolate the cause of the issue.



TCP/IP	OSI Model	Protocols
Application Layer	Application Layer	DNS - DHCP - FTP - HTTPS - LDAP - NTP - POP3 - RTP - RTSP - SSH - SIP - SMTP - Telnet - TFTP
	Presentation Layer	JPEG - MIDI - MPEG - PICT - TIFF
	Session Layer	NetBIOS - NFS - PAP - SCP - SQL - ZIP
Transport Layer	Transport Layer	TCP - UDP
Internet Layer	Network Layer	ICMP - IGMP - IPsec - IPv4 - IPv6 - IPX - RIP
Link Layer	Data Link Layer	ARP - ATM - CDP - FDDI - Frame Relay - HDLC - MPLS - PPP - STP - Token Ring
	Physical Layer	Bluetooth - Ethernet - DSL - ISDN - 802.11 - WiFi

IP Address Classes

Class	Purpose	High-Order Bits	Address Range	Max Number of Hosts
A	Large networks	0	1 to 126	16777,214
B	Medium-size networks	10	128 to 191	65,534
C	Small networks	110	192 to 223	254
D	Multicast	1110	224 to 239	-
E	Experimental	1111	240 to 254	-

Loopback and Private Addresses

Loopback network addresses are used for testing and troubleshooting. Private addresses are reserved for use in private networks and are not routable on the internet.

Loopback Address Range

The address range 127.0.0.1 to 127.255.255.255 is a loopback network used for testing and troubleshooting. Packets sent to a loopback (or localhost) address such as 127.0.0.1 are immediately routed back to the source host without exiting the source host.

IP Address Classes

Class	Purpose	High-Order Bits	Address Range	Max Number of Hosts
A	Large networks	0	1 to 126	16777,214
B	Medium-size networks	10	128 to 191	65,534
C	Small networks	110	192 to 223	254
D	Multicast	1110	224 to 239	-
E	Experimental	1111	240 to 254	-

Private Address Ranges

Several IPv4 address ranges are reserved for use in private networks and are not routable on the internet, including:

- 10.0.0.0–10.255.255.255 (Class A)
- 172.16.0.0–172.31.255.255 (Class B)
- 192.168.0.0–192.168.255.255 (Class C)

Subnet Mask

A subnet mask is a number that hides the network portion of an IPv4 address, leaving only the host portion of the IP address. The network portion of a subnet mask is represented by contiguous “on” (1) bits beginning with the most significant bit.

Physical Layer

When you use a bottom-up approach to diagnose connectivity problems, you begin at the Physical layer of the OSI model by verifying network connections and device availability.

For example, a wireless device may have power to the antenna or transceiver temporarily turned off. A wireless access point may have lost power because a circuit breaker was tripped offline or a fuse was blown. Similarly, a network cable connection may be loose, or the cable may be damaged.

Thus, before you begin inspecting service architectures, you should start with the basics: Confirm physical connectivity.

Data Link Layer

Moving up to the Data Link layer, you verify data link architectures, such as compatibility with a particular standard or frame type.

Although Ethernet is a predominant LAN network standard, devices that roam (such as wireless devices) sometimes automatically switch between Wi-Fi, Bluetooth, and Ethernet networks. Wireless networks usually have specified encryption standards and keys. Connectivity may be lost because a network device or service has been restored to a previous setting and the device is not responding to endpoint requests that are using different settings.

Firewalls and other security policies may also be interfering with connection requests. You should never disable firewalls, but in a controlled network environment with proper procedures established, you may find that temporarily disabling or bypassing a security appliance resolves a connectivity issue. The remedy then is to properly configure security services to allow the required connections.

Network Layer

Various connectivity problems may also occur at the Network layer.
Important troubleshooting steps include confirming proper network names and addresses. Devices may have improperly assigned IP addresses that are causing routing issues or IP address conflicts on the network. A device may have an improperly configured IP address because it cannot communicate with a DHCP server on the network.

Similarly, networks have different identities, such as wireless SSIDs, domain names, and workgroup names. Another common problem exists when a particular network has conflicting names or addresses. Issues with DNS name resolvers may be caused by DNS caching services or connection to the wrong DNS servers. Internet Control Message Protocol (ICMP) is used for network control and diagnostics at the Network layer of the OSI model. Commonly used ICMP commands include ping and traceroute.

These two simple but powerful commands (and other ICMP commands and options) are some of the most commonly used tools for troubleshooting network connectivity issues. You can run ICMP commands in the command line interface on computers, servers, routers, switches, and many other networked devices.

Transport Layer

At the Transport layer, communications are more complex. Latency and network congestion can interfere with communications that depend on timely acknowledgments and handshakes. Time-to-live (TTL) values sometimes have to be extended in the network service architecture to allow for slower response times during peak network traffic hours. Similar congestion problems can occur when new services are added to an existing network or when a local device triggers a prioritized service, such as a backup or an antivirus scan.

Session Layer

Session layer settings can also be responsible for dropped network connections. For example, devices that automatically go into a power standby mode ("sleep") may have expired session tokens that fail when the device attempts to resume connectivity. At the server, failover communications or handshake negotiations with one server may not translate to other clustered servers. Sessions may have to be restarted.

Presentation Layer

Presentation layer conflicts are often related to changes in encryption keys or updates to service architectures that are not supported by various client devices. For example, an older browser may not interoperate with a script or a new encoding standard.

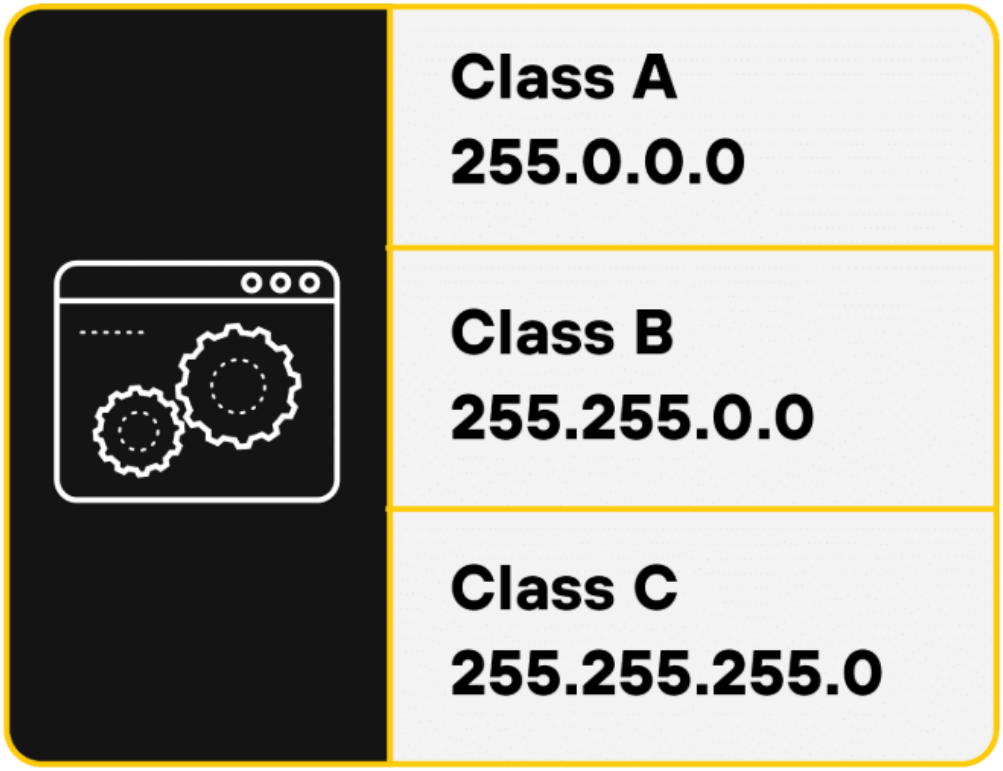
Application Layer

Application layer network connectivity problems are extremely common. Many applications may conflict with other apps. Apps also may have caching or corrupted files that can be remedied only by uninstalling and reinstalling or by updating to a newer version. Some apps also require persistent connections to update services or third parties, and network security settings may prevent those connections from being made.

Subnet Mask

A subnet mask is a number that hides the network portion of an IPv4 address, leaving only the host portion of the IP address. The network portion of a subnet mask is represented by contiguous “on” (1) bits beginning with the most significant bit.

For example, in the subnet mask 255.255.255.0, the first three octets represent the network portion and the last octet represents the host portion of an IP address. The decimal number 255 is represented in binary notation as 11111111. As result, the equivalent of decimal subnet mask 255.255.255.0 in binary notation would be 11111111.11111111.11111111.0.



The default (or standard) subnet masks for Class A, B, and C networks.

IPv4 Network Classes

The system of network classes (A, B, and C) classifies networks by assigning them to specific IP address blocks. This hierarchical structure was designed to make IP address allocation more organized and to streamline network routing and management.

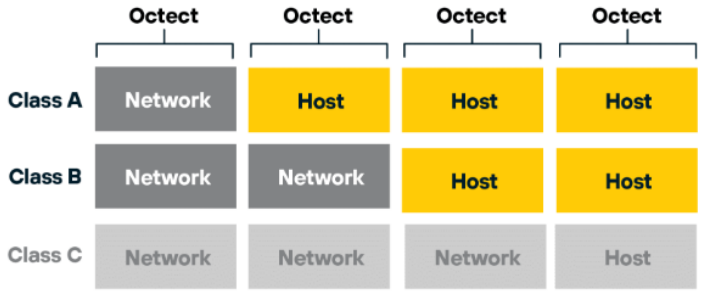
Class A and B Subnets

Class A and B Subnets: Class A and B networks use smaller subnet masks, allowing them to support far more devices (nodes) than Class C.

Class A: Uses an 8-bit default mask (255.0.0.0), providing over 16 million available node addresses.

Class B: Uses a 16-bit default mask (255.255.0.0), providing over 65,000 available node addresses.

Two addresses are always reserved (one for the network, one for broadcast), which is why the total available node count is reduced by two.

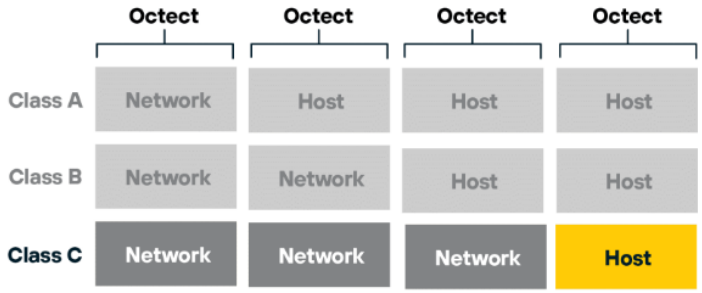


Class C Subnets

A Class C network has 254 usable node (or host) addresses.

It uses a default 24-bit subnet mask (255.255.255.0), which locks in the first three octets as the network portion (e.g., 192.168.1). The last octet is used for devices.

While the last octet provides 256 addresses (2^8), two are reserved: one for the network address and one for the broadcast address. This leaves 254 addresses available for devices.



IPv4 Classless Inter-Domain Routing (CIDR)

IPv4 Classless Inter-Domain Routing (CIDR) is a method for allocating IP addresses and IP routing that replaces classful IP addressing (for example, Class A, B, and C networks) with classless IP addressing.

Variable-Length Subnet Masking

Unlike subnetting, which divides an IPv4 address along an arbitrary (default) classful 8-bit boundary (8 bits for a Class A network, 16 bits for a Class B network, 24 bits for a Class C network), CIDR allocates address space on any address bit boundary (known as variable-length subnet masking, or VLSM).

For example, using CIDR, a network could be assigned one of the following:

- A 24-bit mask (255.255.255.0, instead of the default 8-bit 255.0.0.0 mask) to limit the subnet to only 254 addresses
- A 23-bit mask (255.255.254.0) to limit the subnet to 510 addresses.

Here is an example of variable-length subnet masking.

Supernetting

CIDR is used to reduce the size of routing tables on internet routers by aggregating multiple contiguous network prefixes (known as supernetting), and it also helps slow the depletion of public IPv4 addresses.

For example, using supernetting, instead of advertising 8 separate network routes in a router's routing table:

192.168.2.0/25
192.168.2.0/26
192.168.2.0/27
192.168.2.0/28
192.168.2.240.30
192.168.2.244.30
192.168.2.248.30
192.168.2.252.30

You could instead advertise just one network route (192.168.2.0/24) in a routig table because the 8 separate networks are contained within the "supernetted" 192.168.2.0/24 network route.

Here is an example of supernetting.

Layer 2 Tunneling Protocol (L2TP)

L2TP is supported by most operating systems (including mobile devices). Although L2TP provides no encryption by itself, it is considered secure when used with IPsec.

Secure Socket Tunneling Protocol (SSTP)

Microsoft created SSTP as a VPN tunnel for transporting PPP or L2TP traffic through an SSL 3.0 channel. SSTP is primarily used for secure remote client VPN access, rather than for site-to-site VPN tunnels.

OpenVPN

OpenVPN is a highly secure, open-source VPN implementation that uses SSL/TLS encryption for key exchange. OpenVPN uses up to 256-bit encryption and can run over TCP or UDP. Although OpenVPN is not natively supported by most major operating systems, it has been ported to most major operating systems, including mobile device operating systems.

Microsoft Point-to-Point Encryption (MPPE)

MPPE encrypts data in PPP-based dial-up connections and PPTP VPN connections. MPPE uses the RSA RC4 encryption algorithm to provide data confidentiality and supports 40-bit and 128-bit session keys.

Point-to-Point Tunneling Protocol (PPTP)

PPTP is a basic VPN protocol that uses TCP port 1723 to establish communication with the VPN peer. PPTP then creates a Generic Routing Encapsulation (GRE) tunnel that transports encapsulated Point-to-Point Protocol (PPP) packets between the VPN peers.



Internet Protocol Security (IPsec)

IPsec is a secure communications protocol that authenticates and encrypts IP packets in a communication session. An IPsec VPN requires compatible VPN client software to be installed on the endpoint device. A group password or key is required for configuration. Client-server IPsec VPNs typically require user action to initiate the connection, such as launching the client software and logging in with a username and password.

Click the tabs for more information about configuring and establishing secure communication with IPsec VPNs.

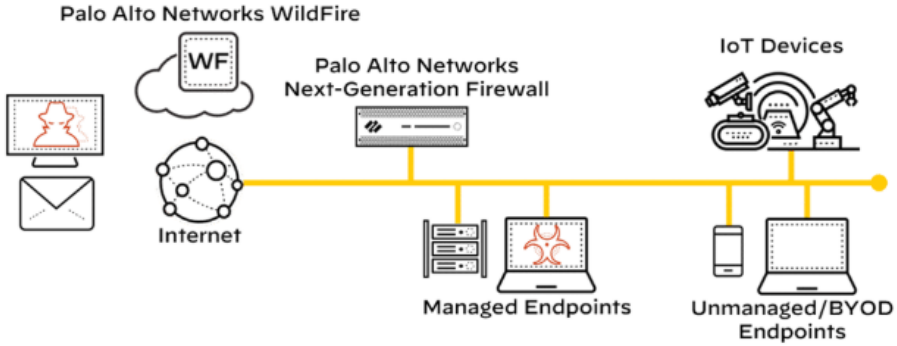
Endpoint security requires implementations of malware analysis and prevention, host intrusion preventions systems - HIPS as well as patch and configuration management.

- Endpoint protection
- Anti-malware and anti-spyware solutions
- Personal firewalls
- HIPS
- MDM
- Server management



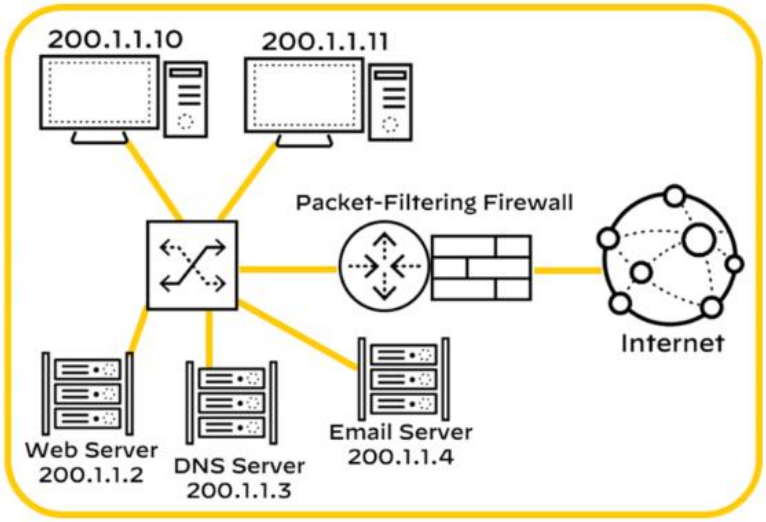
Endpoint Protection Challenges

Advanced malware and script-based attacks can bypass traditional antivirus solutions with ease and potentially wreak havoc on your business.



Operation

Packet filtering firewalls operate up to Layer 4 (Transport layer) of the OSI model and inspect individual packet headers to determine source and destination IP address, protocol (TCP, UDP, ICMP), and port number.



Match

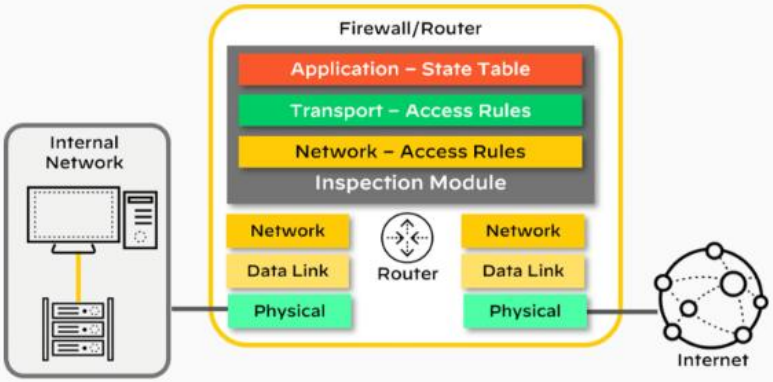
Packet filtering firewalls match source and destination IP address, protocol, and port number information contained within each packet header to a corresponding rule on the firewall that designates whether the packet should be allowed, blocked, or dropped.

Inspection

Packet filtering firewalls inspect and handle each packet individually, with no information about context or session.

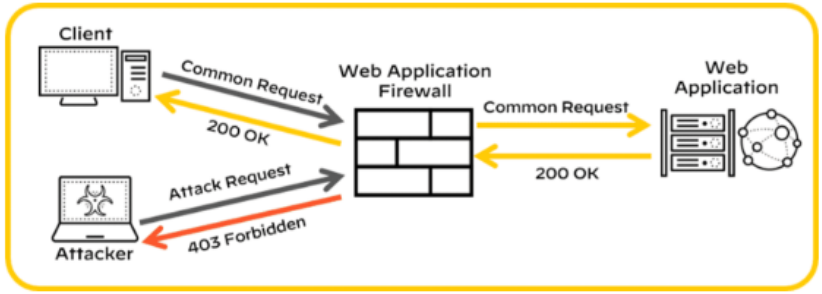
Stateful Packet Inspection Firewalls

Second-generation stateful packet inspection (also known as dynamic packet filtering) firewalls are fast. However, they are port-based and highly dependent on the trustworthiness of the two hosts, because individual packets aren't inspected after the connection is established.



Application Firewalls

Third-generation application firewalls are also known as application-layer gateways, proxy-based firewalls, and reverse-proxy firewalls. Application firewalls operate up to Layer 7 (the application layer) of the OSI model and control access to specific applications and services on the network. These firewalls proxy network traffic rather than permit direct communication between hosts. Requests are sent from the originating host to a proxy server, which analyzes the contents of the data packets and, if the request is permitted, sends a copy of the original data packets to the destination host.



Application firewalls inspect application-layer traffic, so they can identify and block specified content, malware, exploits, websites, and applications or services that use hiding techniques such as encryption and non-standard ports. Proxy servers can also be used to implement strong user authentication and web application filtering and to mask the internal network from untrusted networks. However, proxy servers have a significant negative impact on the overall performance of the network.

Deployment

Organizations deploy next-generation firewalls at the network perimeter and inside the network at logical trust boundaries. All traffic crossing the firewall undergoes a full-stack, single-pass inspection, which provides the complete context of the application, associated content, and user identity. With this level of context, you can align security with your key business initiatives.

Zero Trust Architecture

The next-generation firewall functions as a segmentation gateway in a Zero Trust architecture. By creating a micro-perimeter, the firewall ensures that only known, allowed traffic and legitimate applications have access to the protected surface area.

Next-generation firewalls include many capabilities that enable complete visibility of application traffic flows, including user identity and content. The firewall protects against known attacks, unknown attacks, and advanced persistent threats.

Network Firewalls

Most traditional port-based network firewalls do little to protect endpoints inside the enterprise network from threats that originate from within the network, such as another device that has been compromised by malware and is propagating throughout the network.

Host-Based Firewalls

Personal (or host-based) firewalls are commonly installed and configured on laptop and desktop PCs. Personal firewalls typically operate as Layer 7 (Application layer - OSI Model) firewalls that allow or block traffic based on an individual (or group) security policy. Personal firewalls are particularly helpful on laptops used by remote or traveling users who connect their laptop computers directly to the internet (for example, over a public Wi-Fi connection).

Also, a personal firewall can control outbound traffic from the endpoint to help prevent the spread of malware from that endpoint. However, note that disabling or otherwise bypassing a personal firewall is a common and basic objective in most advanced malware today.

Operating System Firewalls

Windows Firewall is an example of a personal firewall that is installed as part of the Windows desktop or mobile operating system. A personal firewall protects only the endpoint device that it is installed on, but it provides an extra layer of protection inside the network.

Netfilter, or iptables, is the most popular open source, command line interface-based Linux firewall. Many system administrators prefer to use it for their server protection as it operates as the first line of defense for Linux server protection.

Host-Based Intrusion Prevention Systems (HIPS)

A HIPS can be either signature-based or anomaly-based, making it susceptible to the same issues as other signature- and anomaly-based endpoint protection approaches.
Also, HIPS software often causes significant performance degradation on endpoints. A recent Palo Alto Networks survey found that 25 percent of respondents indicated that HIPS solutions "caused significant end user performance impact."

Classifications

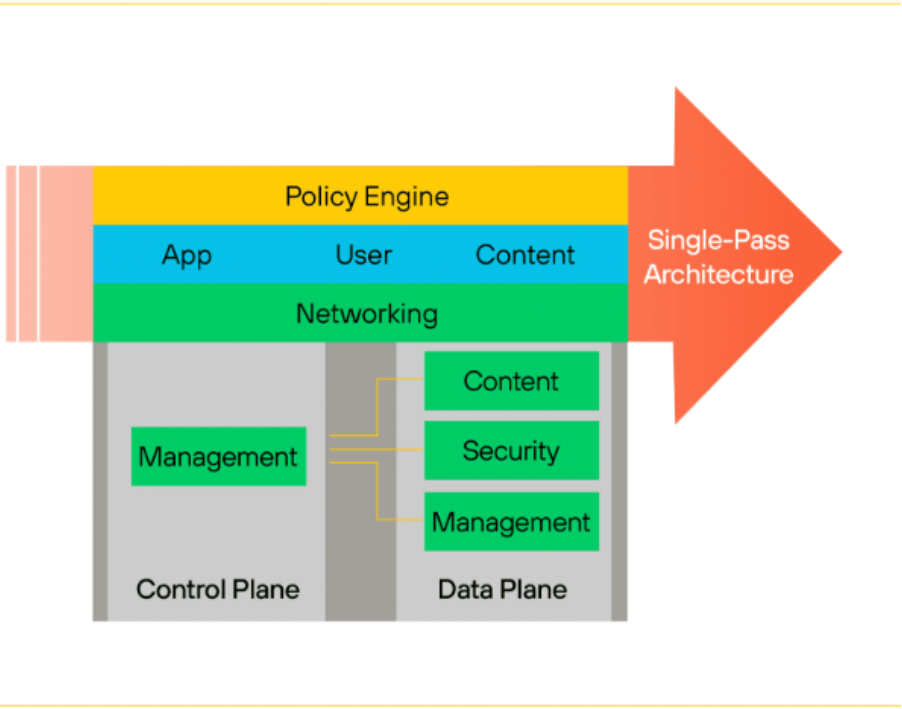
IDSs and IPSs can also be classified as knowledge-based (signature-based) or behavior-based (statistical-anomaly-based) systems. *Flip the cards for more information about knowledge-based and behavior-based systems.*

A knowledge-based system uses a database of known vulnerabilities and attack profiles to identify intrusion attempts. Knowledge-based systems have lower false-alarm rates than behavior-based systems. But to be effective, knowledge based systems must be continually updated with new attack signatures.

A behavior-based system uses a baseline of normal network activity to identify unusual patterns or levels of network activity that might indicate an intrusion attempt. Behavior-based systems are better at detecting new attacks against unknown vulnerabilities. But behavior-based systems have a much higher false-positive rate than knowledge-based systems.

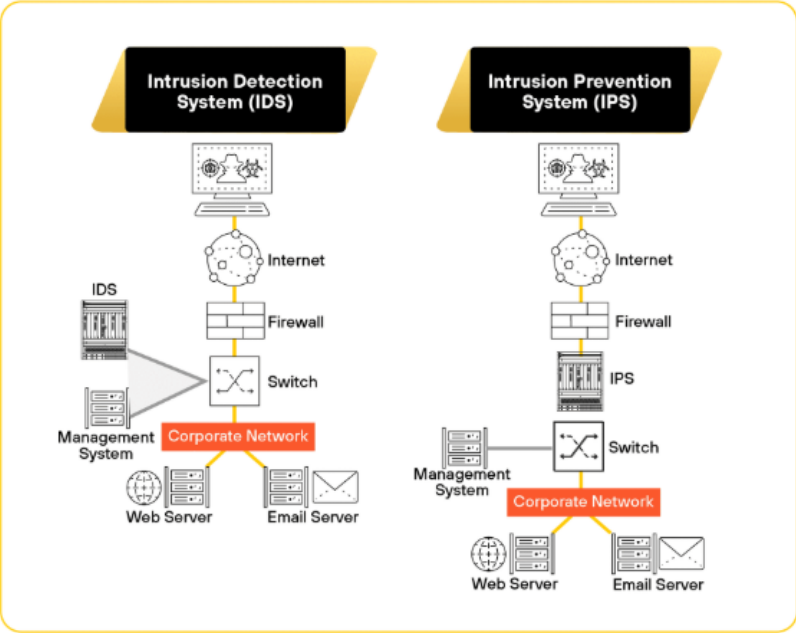
Single-Pass Architecture

A single-pass architecture is a unique integration of software and hardware that simplifies management, streamlines processing, and maximizes performance. The single-pass architecture integrates multiple threat prevention disciplines (IPS, anti-malware, URL filtering, etc.) into a single stream-based engine with a uniform signature format. This architecture allows traffic to be fully analyzed in a single pass without the performance degradation seen in multifunction gateways. The software is tied directly to a parallel processing hardware platform that uses function-specific processors for threat prevention to maximize throughput and minimize latency.



Single Stream-Based Engine

The use of one common engine realizes two key benefits. First, unlike file proxies that need to download the entire file before they can scan the traffic, a stream-based engine scans traffic in real time, only reassembling packets as needed and only in very small amounts. Second, unlike traditional approaches, all traffic can be scanned with a single engine, instead of multiple scanning engines.



Distinctive IDS & IPS Features

Unlike traditional packet filtering firewalls and stateful packet inspection firewalls, which examine only packet header information, IDSs and IPSs examine both the packet header and the payload of network traffic. IDSs and IPSs attempt to match known-bad, or malicious, patterns (or signatures) in inspected packets. IDSs and IPSs are typically deployed to detect and block exploits of software vulnerabilities on target networks.

Intrusion Detection System (IDS)

Description and functionality:

- Is considered a passive system
- Monitors and analyzes network activity
- Sends alerts about potential attacks and vulnerabilities on the network

Disadvantages:

- Doesn't perform any preventive action to stop an attack

Intrusion Prevention System (IPS)

Description and functionality:

- Is considered an active system
- Performs all of the functions of an IDS
- Automatically blocks or drops suspicious pattern-matching activity on the network in real time

Disadvantages:

- Must be placed inline along a network boundary and is thus directly susceptible to attack itself
- Can trigger false alarms that inadvertently block authorized users and applications and must be properly identified and filtered
    - A false positive occurs when legitimate traffic is improperly identified as malicious traffic.
- Can be used to deploy a denial-of-service (DoS) attack by flooding the IPS, thus blocking connections until no connection or bandwidth is available

Networking Knowledge

In the world of cyber security, having a strong foundation in networking knowledge is crucial. It's important to understand the fundamental concepts and mechanisms that govern how data is transferred, communicated, and secured across digital networks.

Topics

- **Network Architecture**: Learn about the different networking models, such as the OSI model and TCP/IP model, which define how data is structured, transmitted, and received in a network.
- **Network Protocols**: Familiarize yourself with various network protocols that are essential for effective communication between devices, including HTTP, HTTPS, FTP, and more. These protocols ensure that data is transmitted reliably and securely across networks.
- **IP Addressing and Subnetting**: Gain an understanding of IP addresses (both IPv4 and IPv6), how they are assigned, and how subnetting works to divide networks into smaller segments for better management and security.
- **Routing and Switching**: Learn about the roles of routers and switches in a network, as well as related technologies and protocols like DHCP, NAT, and various routing protocols (such as OSPF and BGP).
- **Wireless Networking**: Delve into the world of wireless networks by studying the different types of wireless technologies like Wi-Fi, Bluetooth, and cellular networks. Understand the security concerns and best practices associated with wireless communication.
- **Network Security**: Explore various techniques and tools used to defend networks from cyber threats, including firewalls, intrusion detection systems (IDS), intrusion prevention systems (IPS), and VPNs. Learn about security protocols like SSL/TLS, encryption algorithms, and secure access control mechanisms.
- **Network Troubleshooting**: Understand common network issues and how to resolve them, using various network troubleshooting tools and methodologies like ping, traceroute, and Wireshark.

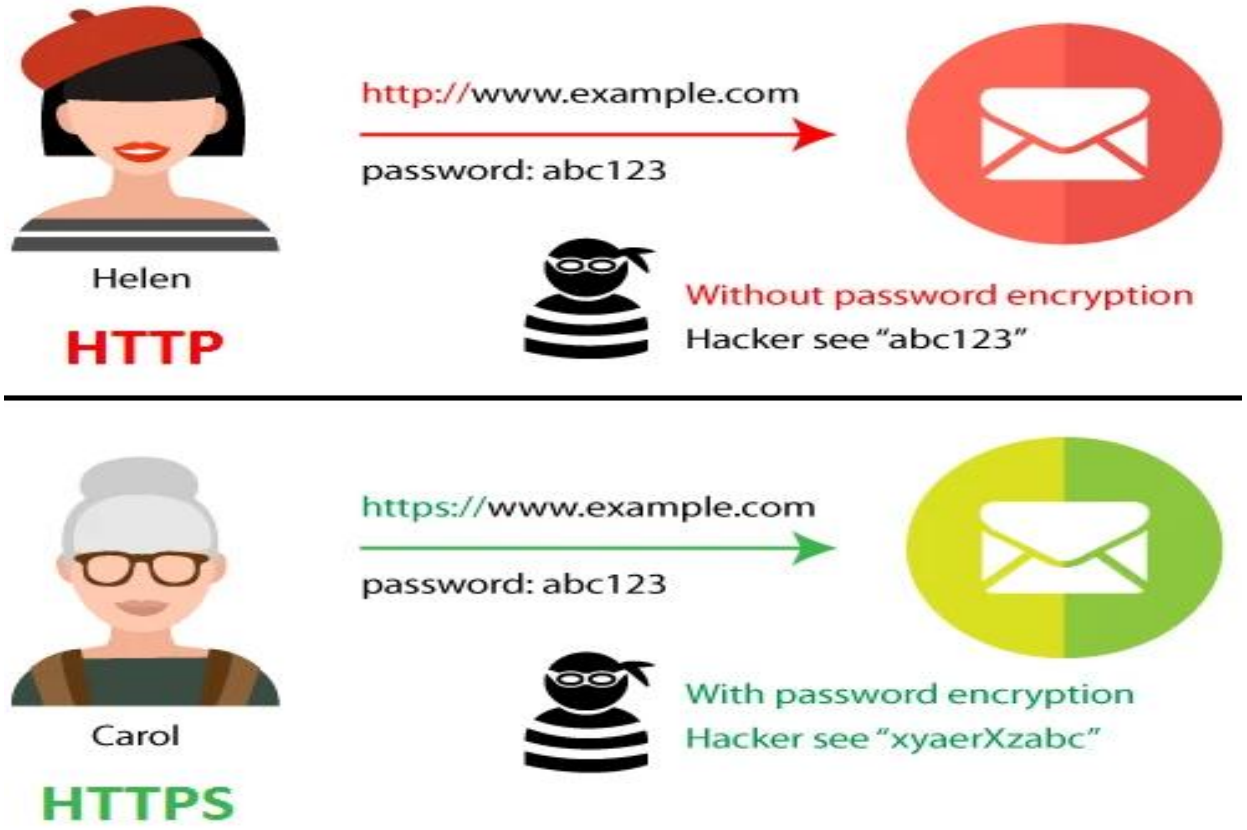
Common Protocols and their Uses

In this section, we will discuss some of the most common protocols used in networking and their importance in maintaining cyber security. Protocols are a set of rules and procedures that define how data should be transmitted, formatted, and processed over a network.

HyperText Transfer Protocol (HTTP) and HTTPS

HTTP, or HyperText Transfer Protocol, is the foundation of data communication on the World Wide Web. It defines how data should be formatted and transmitted between a client (like your browser) and a web server. HTTP is a stateless protocol, meaning each request and response pair is independent from others.

HTTPS, or HTTP Secure, is a secure version of HTTP that encrypts data between the client and server using Secure Sockets Layer (SSL) or Transport Layer Security (TLS) to protect sensitive data from being intercepted or tampered with.



Internet Protocol (IP)

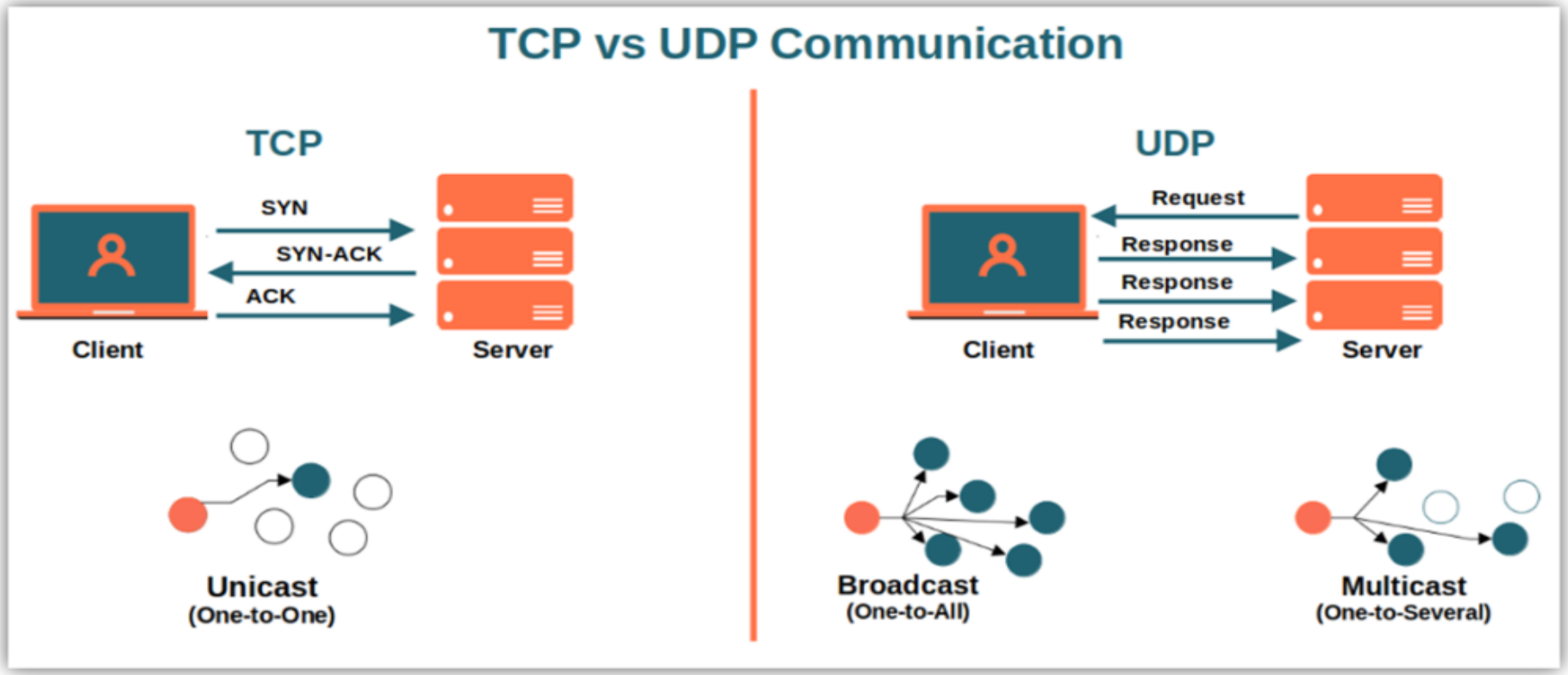
Internet Protocol (IP) is responsible for delivering packets from the source host to the destination host based on their IP addresses. IP is the primary protocol in the Internet Layer of the Internet Protocol Suite and has two main versions - IPv4 and IPv6.

Transmission Control Protocol (TCP)

TCP, or Transmission Control Protocol, is a reliable, connection-oriented protocol that ensures data is delivered correctly between applications over a network. It ensures accurate and complete data delivery by establishing a connection, segmenting data into smaller packets, verifying the receipt of packets, and reordering packets to their original sequence.

User Datagram Protocol (UDP)

UDP, or User Datagram Protocol, is a connectionless communication protocol used for fast and efficient data transmission. Unlike TCP, UDP does not provide error checking or guarantee delivery, making it suitable for real-time applications like video streaming and online gaming where low latency is crucial.



Domain Name System (DNS)

The Domain Name System (DNS) is responsible for translating human-readable domain names (like www.example.com) into corresponding IP addresses that computers understand. This process is called domain name resolution. DNS is an essential component of internet communication, as it allows users to access websites using easy-to-remember names instead of numerical IP addresses.

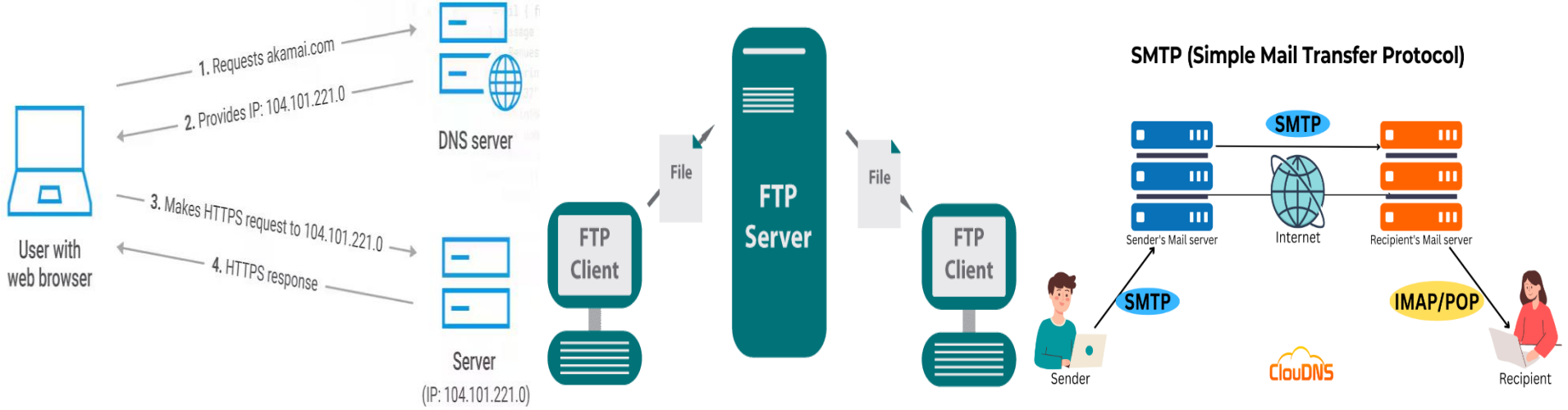
File Transfer Protocol (FTP)

File Transfer Protocol (FTP) is a standard network protocol used for transferring files from one host to another over a TCP-based network, such as the Internet. FTP is commonly used for sharing files and transferring files between a client and a server.

Simple Mail Transfer Protocol (SMTP)

Simple Mail Transfer Protocol (SMTP) is the standard protocol for sending email messages across a network. It defines how email messages should be formatted, encrypted, and relayed between email clients, servers, and other email systems.

Understanding these common protocols and their roles in network communication is vital for ensuring the proper implementation of cyber security measures. It will help you better identify potential vulnerabilities and make informed decisions on network defense strategies.



Transmission Control Protocol (TCP) Ports

- **FTP (File Transfer Protocol) - Ports 20 and 21**: FTP is a widely used protocol for transferring files.
- **SSH (Secure Shell) - Port 22**: SSH allows secure communication and remote access to devices over an unsecured network.
- **Telnet - Port 23**: Telnet is a text-based protocol that allows you to interact with remote devices over networks.
- **SMTP (Simple Mail Transfer Protocol) - Port 25**: SMTP is a protocol for sending and receiving emails.
- **DNS (Domain Name System) - Port 53**: DNS translates human-readable domain names into IP addresses to facilitate communication between devices.
- **HTTP (Hypertext Transfer Protocol) - Port 80**: HTTP is the primary protocol used for communication on the World Wide Web.
- **POP3 (Post Office Protocol 3) - Port 110**: POP3 is a protocol for receiving emails from your email server.
- **IMAP (Internet Message Access Protocol) - Port 143**: IMAP is a more advanced email protocol that allows you to access and manage your emails on the email server.
- **HTTPS (Hypertext Transfer Protocol Secure) - Port 443**: HTTPS is an encrypted and secure version of HTTP.
- **RDP (Remote Desktop Protocol) - Port 3389**: RDP is a Microsoft-developed protocol for remotely accessing Windows devices.

User Datagram Protocol (UDP) Ports

- **DHCP (Dynamic Host Configuration Protocol) - Ports 67 and 68**: DHCP is used to allocate IP addresses to devices within a network.
- **DNS (Domain Name System) - Port 53**: (same function as in TCP)
- **TFTP (Trivial File Transfer Protocol) - Port 69**: TFTP is a simplified version of FTP for quick and easy file transfer.
- **SNMP (Simple Network Management Protocol) - Port 161**: SNMP enables monitoring and managing network devices, including printers, routers, and switches.
- **NTP (Network Time Protocol) - Port 123**: NTP is a standard protocol used to synchronize time across network devices.

SSL and TLS Basics

Secure Sockets Layer (SSL) and Transport Layer Security (TLS) are cryptographic protocols designed to provide secure communication over a computer network. They play a vital role in protecting sensitive information transmitted online, such as login credentials, financial information, and private user data.

Secure Sockets Layer (SSL)

SSL is the predecessor to TLS and was first introduced in the 1990s. It creates an encrypted connection between a client (typically a web browser) and a server to ensure that any data transmitted remains private and secure. SSL uses a combination of symmetric and asymmetric encryption methods, as well as digital certificates, to establish and maintain secure communication.

Transport Layer Security (TLS)

TLS is an improved and more secure version of SSL, with TLS 1.0 being released as an upgrade to SSL 3.0. The current version, as of this guide, is TLS 1.3. TLS provides a more robust and flexible security framework, addressing many of the vulnerabilities present in SSL. While many people still refer to SSL when discussing secure web communication, it's important to note that SSL has been deprecated, and TLS is the best-practice standard for secure communication.

Key Components

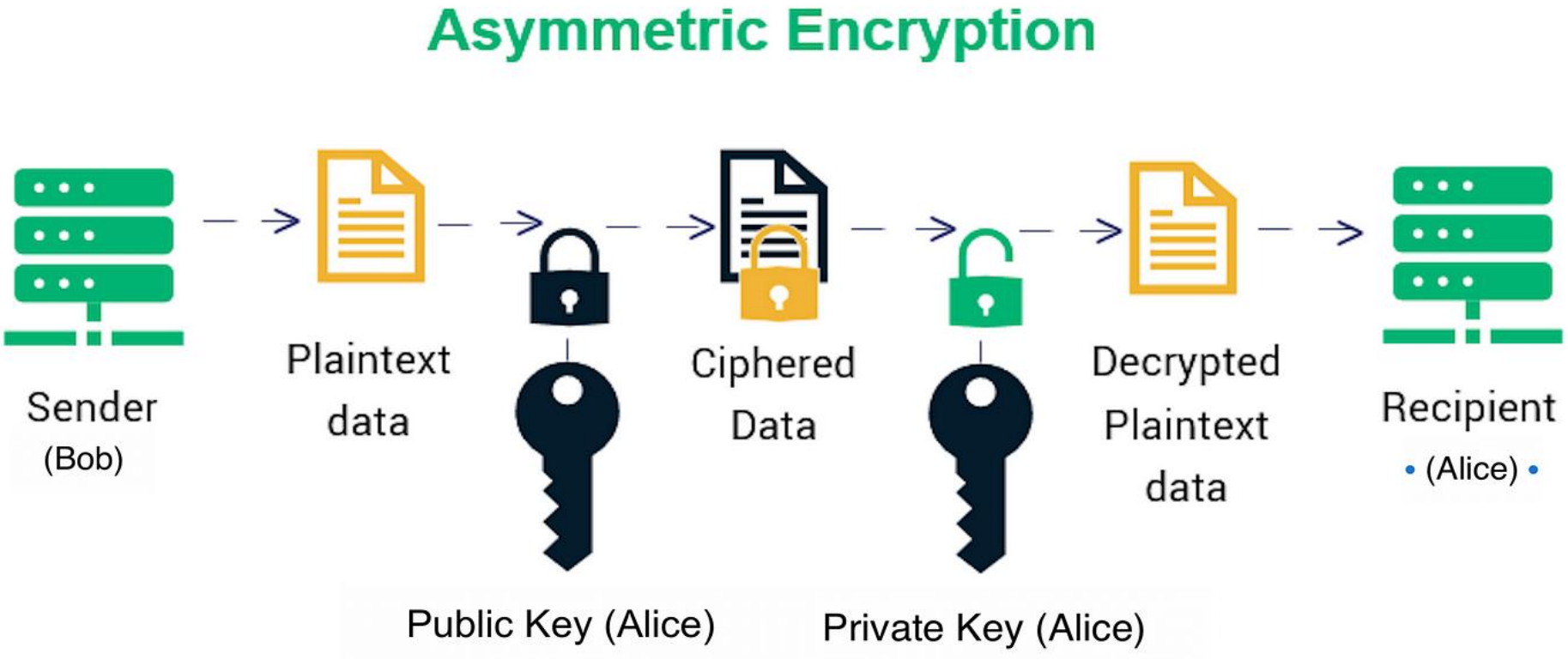
- **Encryption**: SSL and TLS use powerful algorithms to protect data through encryption, ensuring it's unreadable by anyone without the proper decryption keys.
- **Authentication**: SSL/TLS digital certificates verify the identities of clients and servers, providing trust and authenticity.
- **Integrity**: These security protocols use message authentication codes to ensure that the data sent between clients and servers has not been tampered with during transmission.

-

Handshake Process

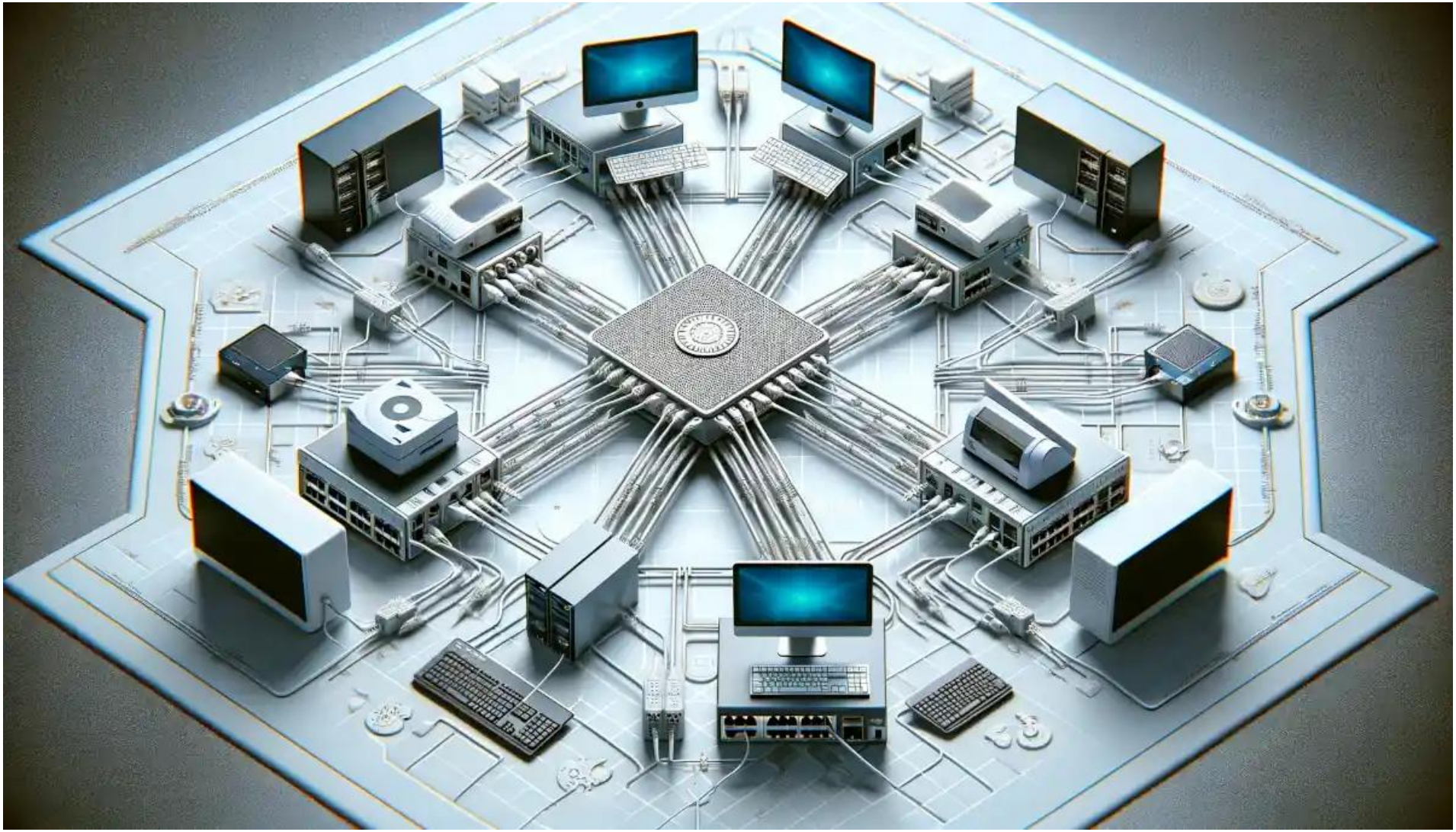
SSL and TLS follow a series of steps, known as the “handshake process,” to create a secure connection:

- **Client hello**: The client initiates the handshake process by sending a message with supported cryptographic algorithms, random numbers, and session information.
- **Server hello**: The server responds with its chosen cryptographic algorithms, random numbers, and its digital certificate. Optionally, the server can request the client’s certificate for mutual authentication.
- **Client verification**: The client verifies the server’s certificate and may send its own if requested. It then creates a pre-master secret, encrypts it with the server’s public key, and sends it to the server.
- **Key generation and exchange**: Both the client and server generate the master secret and session keys using the pre-master secret and shared random numbers. These keys are used for encrypting and decrypting the data transmitted.
- **Secured connection**: Once the keys are exchanged, the client and server can now communicate securely using the established encryption and keys.



Network Topologies

- Network topologies describe the arrangement of various devices in a network, their connections, and the flow of data between them. Understanding common network topologies can help you identify potential vulnerabilities and enhance your overall cybersecurity posture. Here, we'll briefly discuss the different types of network topologies and their advantages and disadvantages.



Bus Topology

In a bus topology, all devices in the network are connected to a single communication medium (usually a coaxial cable) called a “bus.” Data is transmitted in a single direction along the bus, and devices look for their address in the data to know if it’s meant for them.

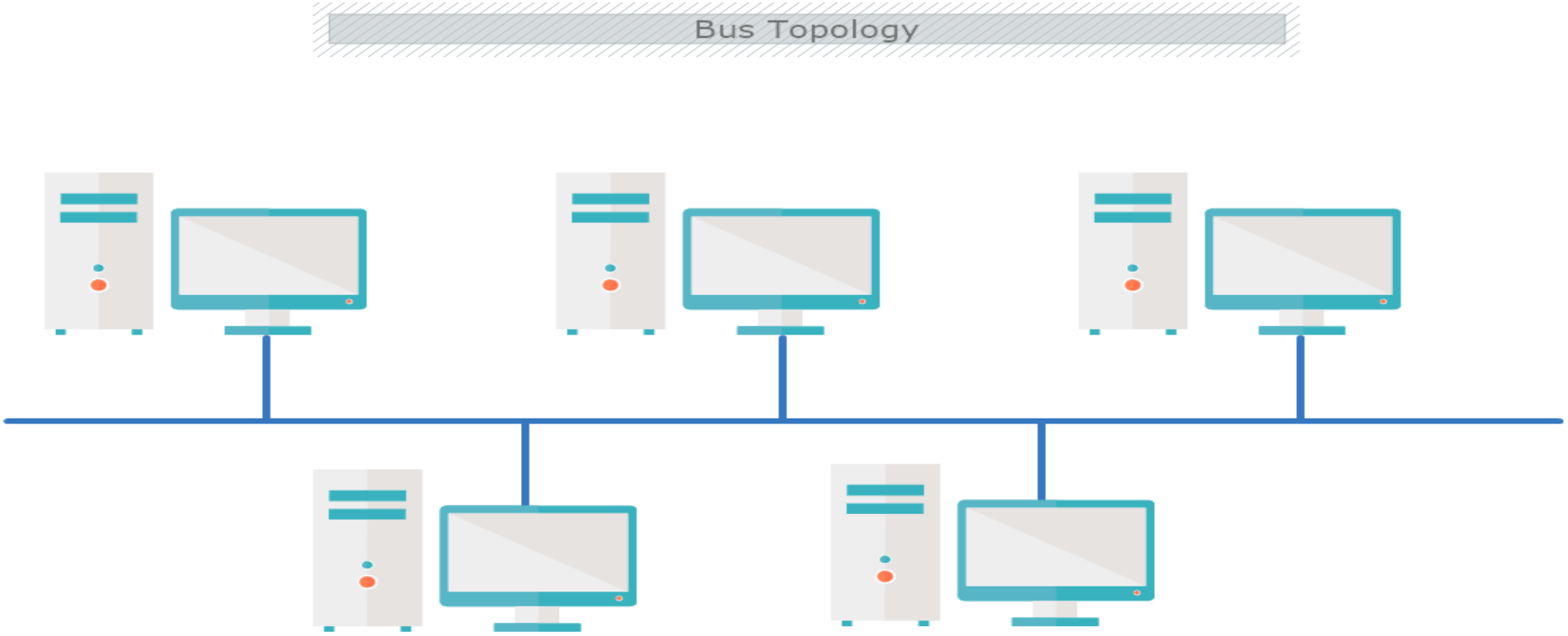
Advantages:

- Easy to set up and extend
- Requires less cabling than other topologies

Disadvantages:

- If the main cable fails, the entire network fails
- Performance degrades as more devices are added

Limited cable length and number of devices



Star Topology

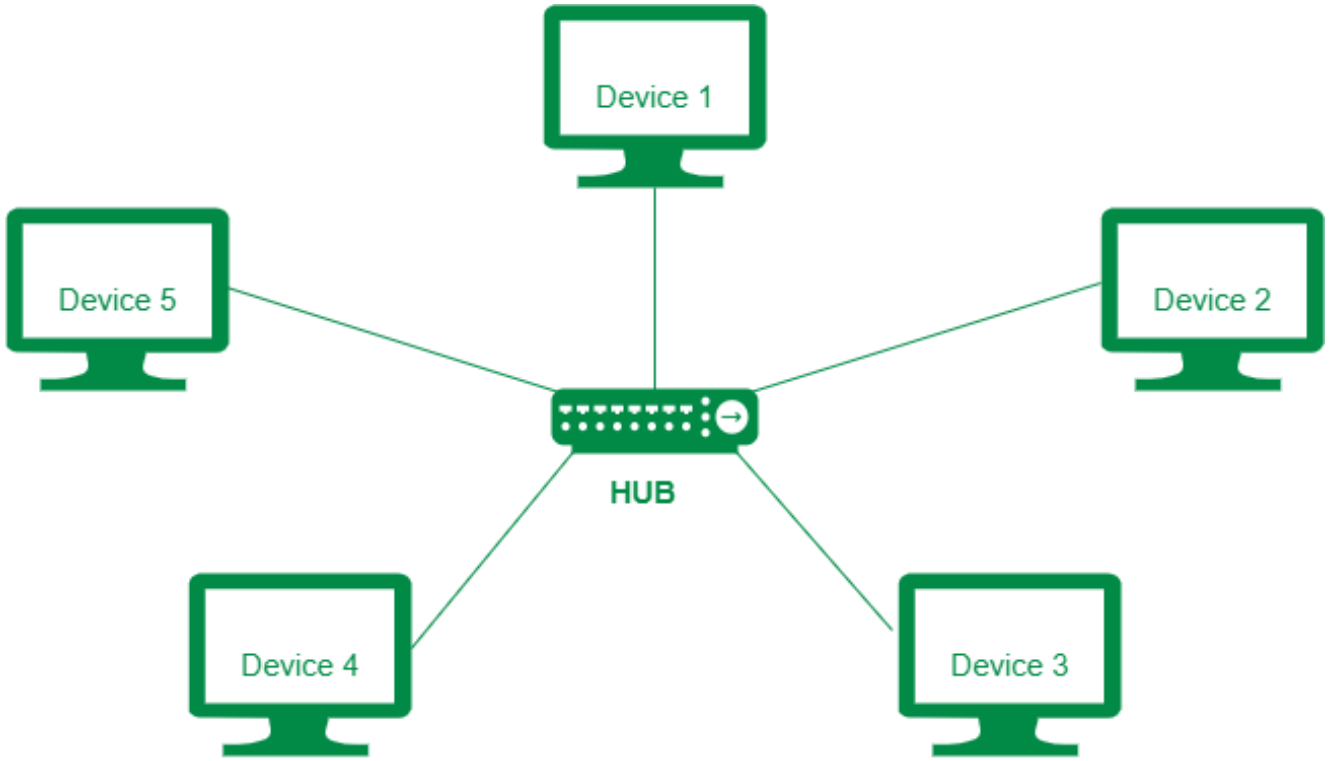
A star topology connects all devices to a central point or hub (typically a switch or a router). The central point is responsible for transmitting data between devices in the network.

Advantages:

- Easy to add or remove devices without affecting the rest of the network
- If one device fails, it doesn't affect the entire network
- Centralized management

Disadvantages:

- Requires more cabling than bus topology
- If the central hub fails, the entire network fails



Star Topology

Ring Topology

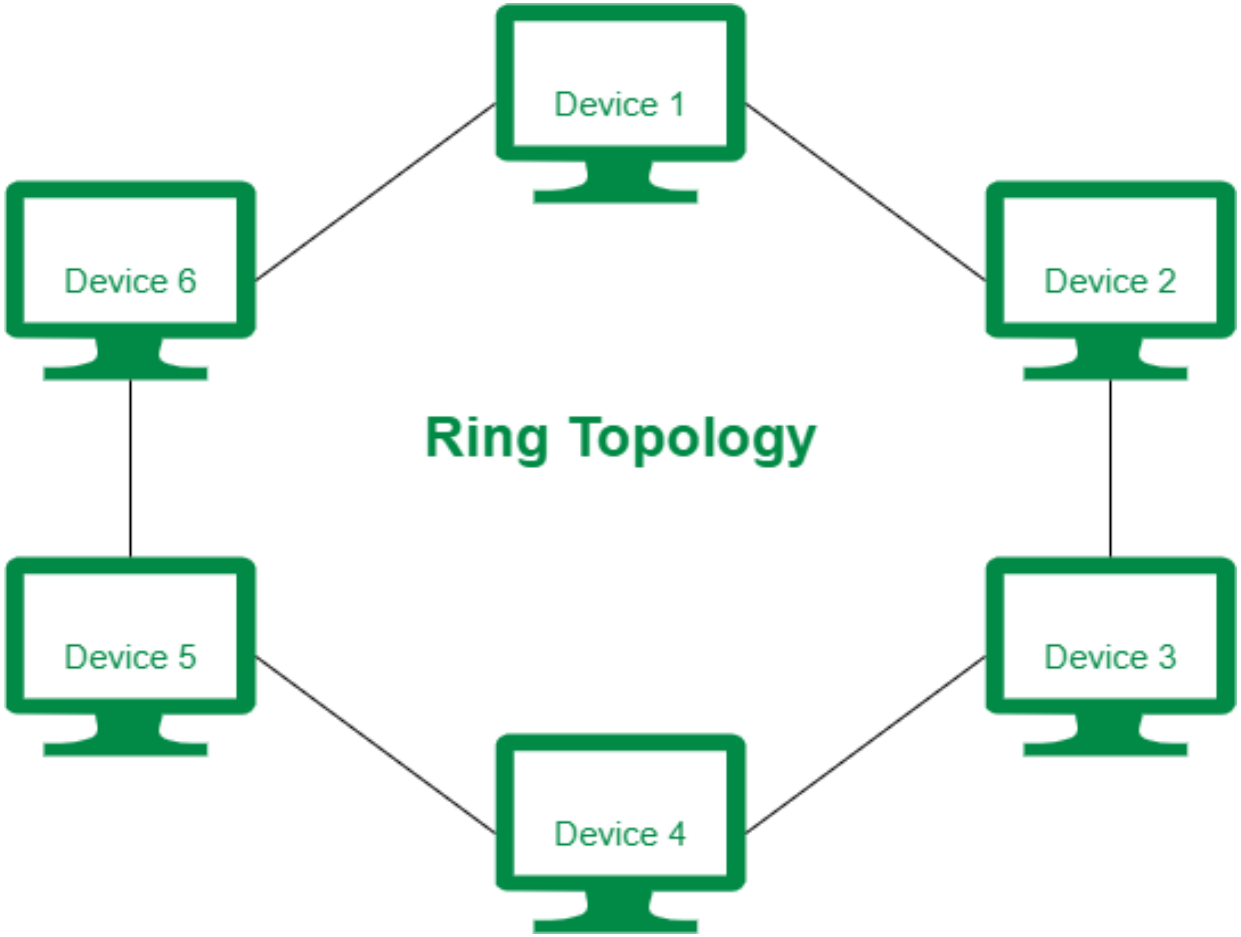
In a ring topology, devices are connected in a circular pattern, with each device having exactly two neighbors. Data is transmitted in one direction around the ring, passing through each device before reaching its destination.

Advantages:

- Equal access to resources for all devices
- Can handle high-traffic loads

Disadvantages:

- Adding or removing devices can disrupt the network
- If one device fails, it can affect the entire network
- Data transmission can be slow due to the loop structure



Mesh Topology

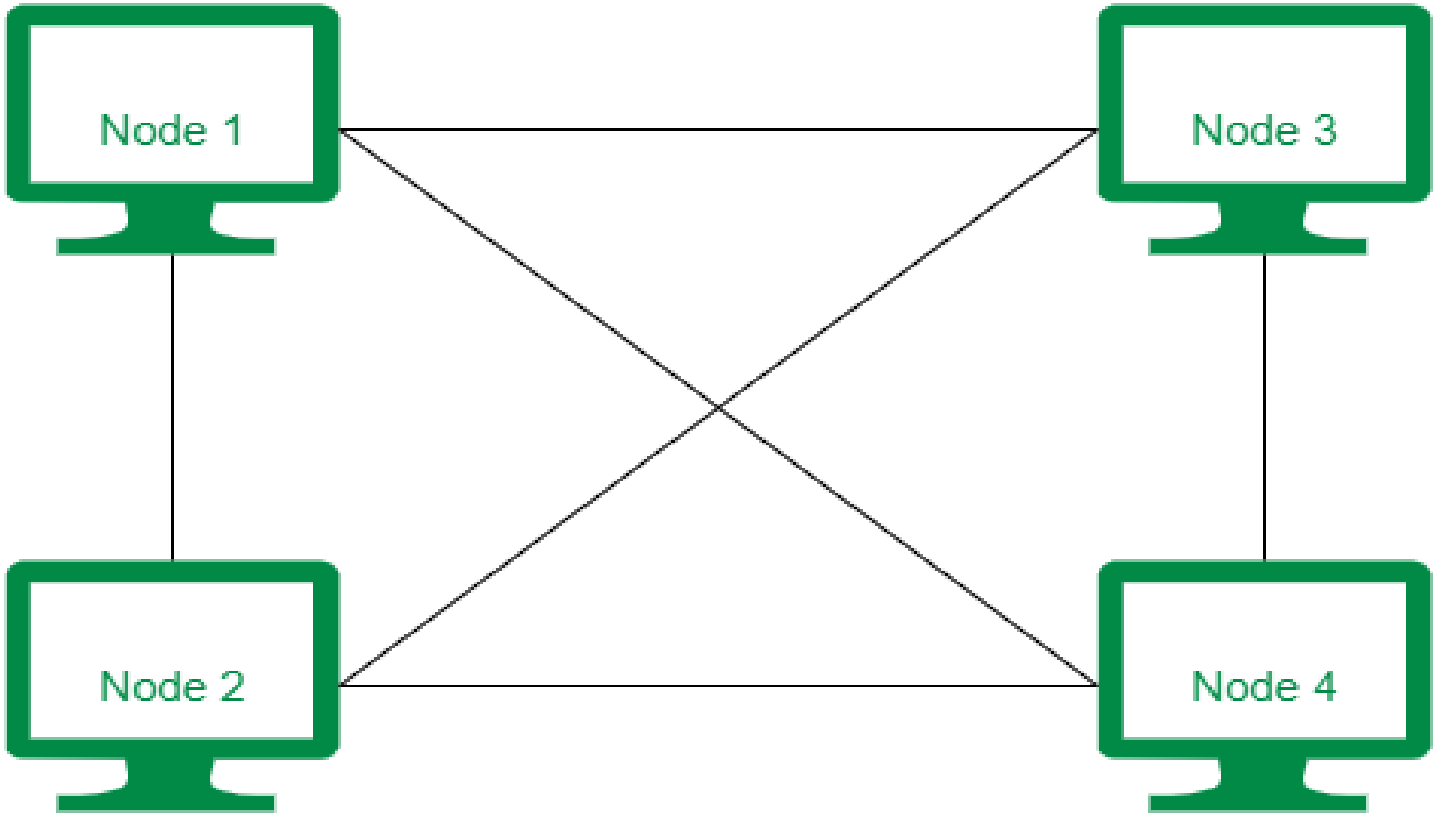
A mesh topology connects all devices directly to every other device in the network. It can be a full mesh (where every device is connected to every other device) or a partial mesh (where some devices are connected to all others, while others maintain only a few connections).

Advantages:

- High fault-tolerance and redundancy, making it more resilient
- Eliminates the need for a central hub

Disadvantages:

- Requires a large number of cables, making it expensive and difficult to manage
- Can be challenging to set up and maintain



Hybrid Topology

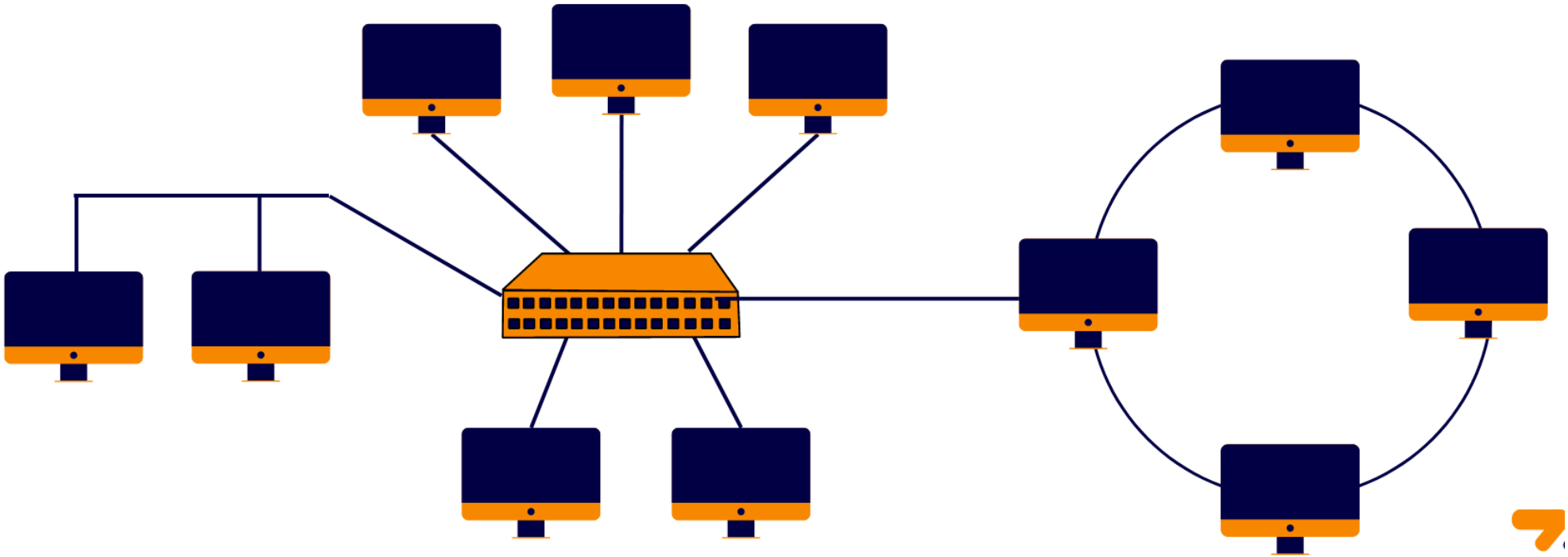
A hybrid topology combines two or more different topologies, such as a star and ring topology, in a single network. It can be customized to fit specific network requirements and performance needs.

Advantages:

- Can be tailored to meet specific needs
- Optimizes the strengths of various topologies

Disadvantages:

- Can be complex and difficult to manage
- More expensive than other topologies



Ping

Ping is a basic command-line tool used to test the reachability of a network host. It sends ICMP Echo Request packets to the target host and waits for an ICMP Echo Reply. If the target host is reachable, you will receive the packets back with round-trip time statistics.
Usage: **ping [target host/IP]**

Host

Host is a basic command-line tool used to reveal final destination/ip of connection.
Usage: **host targetdomain.com**

Traceroute/tracert

traceroute (Linux) and **tracert** (Windows) are command-line tools used to display the path taken by packets across a network. They can help to identify routing problems, latency, and packet loss.
Usage: **traceroute [target host/IP]** or **tracert [target host/IP]**

Nslookup

nslookup is a network administration command-line tool used to query Domain Name System (DNS) servers for host information or IP address resolution.
Usage: **nslookup [hostname]**

Netstat

The **netstat** command is a versatile command-line tool that displays network connections, routing tables, and network interface statistics. It can help identify critical connections, open ports, and listening services.
Usage: **netstat [-options]**

Nmap

Nmap (Network Mapper) is an open-source tool for network discovery and security auditing. It can scan for open ports, running services, and identify network vulnerabilities.
Usage: **nmap [-options] [target host/IP]**

Wireshark

Wireshark is a widely-used network protocol analyzer that allows you to capture and analyze network traffic in real-time. It provides detailed information about packets, protocols, and network behavior that aids in troubleshooting and security analysis.
Download link: https://www.wireshark.org/download.html

Public Key Infrastructure (PKI)

Introduction to PKI

Public Key Infrastructure (PKI) is a framework of technologies, policies, and processes used to manage digital certificates and public-key cryptography. It helps establish trust between users, computers, servers, and applications.

Topics

- **Public-Key Cryptography:** Understanding public keys and private keys.
- **Digital Certificates:** Learning how certificates identify websites, servers, and other entities.
- **Certificate Authority (CA):** Understanding how trusted authorities issue and sign digital certificates.
- **Root and Intermediate Certificates:** Learning how certificate authorities establish a chain of trust.
- **Certificate Chain Validation:** Understanding how a client verifies a server's certificate.
- **Certificate Expiration and Revocation:** Learning why certificates expire and how revoked certificates are handled.
- **PKI in HTTPS:** Understanding how PKI supports authentication and trust in TLS connections.
- **PKI in Cybersecurity:** Exploring how certificates are used in secure communication, device authentication, and digital signatures.

Practical Exercise

Inspect a website's TLS certificate and identify its issuer, subject, validity period, and certificate chain.

Network Address Translation (NAT)

Introduction to NAT

Network Address Translation (NAT) is a networking technique that modifies IP address information in packet headers as traffic passes through a router or firewall. It is commonly used to allow devices with private IP addresses to communicate with external networks.

Topics

- **Private and Public IP Addresses:** Understanding the differences between internal and internet-routable addresses.
- **How NAT Works:** Learning how addresses are translated when packets travel between networks.
- **Static NAT:** Mapping a specific internal address to a specific external address.
- **Dynamic NAT:** Translating internal addresses using a pool of public addresses.
- **Port Address Translation (PAT):** Allowing multiple internal devices to share a public IP address by using different port mappings.
- **Source NAT (SNAT):** Changing the source address of outgoing traffic.
- **Destination NAT (DNAT):** Changing the destination address of incoming traffic.
- **Port Forwarding:** Directing incoming connections to a specified internal service.
- **NAT and Firewalls:** Understanding the difference between address translation and traffic filtering.
- **NAT in Cybersecurity:** Examining exposed services, network boundaries, and common configuration mistakes.

Practical Exercise

Configure NAT in a virtual network and observe how packet source and destination addresses change during communication.

Virtual LAN (VLAN) on Hardware

Introduction to Hardware VLANs

A Virtual Local Area Network (VLAN) is a logical network segment created on a network switch. VLANs allow devices connected to the same physical switching infrastructure to be separated into different Layer 2 broadcast domains.

Topics

- **VLAN Fundamentals:** Understanding the purpose of network segmentation.
- **VLAN IDs:** Learning how VLANs are identified on managed switches.
- **Access Ports:** Configuring switch ports that carry traffic for a single VLAN.
- **Trunk Ports:** Carrying traffic for multiple VLANs over a single connection.
- **IEEE 802.1Q Tagging:** Understanding how VLAN information is added to Ethernet frames.
- **VLAN Assignment:** Assigning switch ports to the appropriate VLANs.
- **Inter-VLAN Routing:** Understanding how devices in different VLANs communicate through a Layer 3 device.
- **VLANs and Network Security:** Separating departments, servers, guest networks, and management interfaces.
- **VLAN Misconfiguration:** Understanding common configuration errors and their security implications.
- **VLAN Hopping:** Introduction to attacks that attempt to cross VLAN boundaries through misconfigured switching environments.

Practical Exercise

Create two VLANs on a managed switch, assign different ports to each VLAN, and test communication between connected devices.

Virtual LAN (VLAN) on Software Servers

Introduction to Software-Based VLANs

VLANs can also be configured and managed through software on supported servers, hypervisors, and virtual networking platforms. This allows virtual machines and server interfaces to participate in logically separated networks without requiring a separate physical switch for every network.

Topics

- **Virtual Networking Fundamentals:** Understanding virtual network interfaces, bridges, and virtual switches.
- **Software-Based Network Segmentation:** Separating virtual machines and services into different logical networks.
- **VLAN Interfaces on Linux:** Introduction to configuring VLAN-tagged network interfaces.
- **Linux Bridges and VLAN Filtering:** Understanding how bridges connect virtual and physical network interfaces while maintaining VLAN separation.
- **Hypervisor Virtual Networking:** Introduction to VLAN configuration for virtual machines.
- **Tagged and Untagged Traffic:** Understanding how VLAN traffic is handled by virtual network interfaces.
- **VLANs on Virtual Machines:** Assigning virtual machines to appropriate VLANs.

- **Virtual Network Isolation:** Limiting unnecessary communication between virtual machines and server services.
- **VLAN Troubleshooting:** Identifying common tagging, bridge, and interface configuration problems.
- **Software VLAN Security:** Understanding why correct configuration and access controls are necessary to maintain isolation.

Practical Exercise

Create two isolated virtual networks on a supported hypervisor or Linux server, connect a virtual machine to each network, and test connectivity and isolation.

Demilitarized Zone (DMZ) in Network Security

Introduction to DMZ

A Demilitarized Zone (DMZ) is a network segment used to host services that must be accessible from untrusted networks, such as the internet. It is designed to separate public-facing systems from an organization's internal network and reduce the risk of unauthorized access to internal resources.

Topics

- **DMZ Fundamentals:** Understanding the purpose of a DMZ and its role in network security.
- **DMZ Architecture:** Exploring common DMZ designs and network boundaries.
- **Single-Firewall DMZ:** Understanding how a firewall can separate the internet, DMZ, and internal network using different interfaces or security zones.
- **Dual-Firewall DMZ:** Understanding how two firewalls can provide separate security boundaries around a DMZ.
- **Public-Facing Services:** Learning why web servers, mail gateways, and reverse proxies may be placed in a DMZ.
- **DMZ and Firewall Rules:** Restricting traffic between the internet, DMZ, and internal network.
- **DMZ and NAT:** Understanding how public IP addresses and destination NAT can expose selected services.
- **DMZ and VLANs:** Using VLANs to implement network separation in suitable network architectures.
- **DMZ Server Hardening:** Reducing exposed services, restricting administrative access, and applying security updates.
- **DMZ Monitoring and Logging:** Monitoring traffic and investigating suspicious activity.
- **DMZ Security Principles:** Applying least privilege, network segmentation, and defense in depth.

Practical Exercise

Build a virtual network with an external test machine, a firewall, a DMZ web server, and an internal network. Configure firewall rules to allow required web traffic while blocking unauthorized connections from the DMZ to the internal network.

Identity Access Management

IAM is important because having a network connection doesn't mean a user should automatically have access to every server, application, or file on that network.

What does IAM control?

1. **Identity** - Who or what is requesting access?
   Examples: employees, administrators, servers, service accounts.
2. **Authentication** - How do we verify the identity?
   Examples: passwords, MFA, security keys, certificates.
3. **Authorization -** What is that identity allowed to access or do?
   Examples: read files, administer a server, manage network devices.
4. **Auditing and accountability** - What actions did the identity perform?
   Examples: login records, permission changes, administrative activity.

How IAM relates to networking

Imagine a company network with three groups:

Identity	Permitted access
Employee	Internal applications and approved shared files
Network administrator	Switches, routers, and network management interfaces
Web server service account	Only the specific systems and resources required by its application

IAM helps enforce these permissions. Network controls, such as firewalls and VLANs, provide additional restrictions on which systems can communicate.

The difference: VLANs control network segmentation, firewalls control network traffic, and IAM controls identities and their permissions. They complement each other.

Important IAM concepts for your cybersecurity syllabus

- RBAC (Role-Based Access Control): Permissions are assigned according to roles.
- Least Privilege: Give users and services only the access they need.
- MFA (Multi-Factor Authentication): Require multiple types of verification.
- AAA (Authentication, Authorization, Accounting): A framework commonly used to control and record access to network devices.
- RADIUS and TACACS+: Protocols used in centralized authentication and access control, particularly for network infrastructure.
- LDAP and Active Directory: Technologies commonly used to manage directory identities and organizational access.
- PAM (Privileged Access Management): Controls and monitors access to powerful administrative accounts.
- SSO (Single Sign-On): Allows users to access multiple integrated applications through one authentication process.

IAM Goals

Click the icons for more information about the goals of IAM.

Compliance

IAM can help organizations comply with regulations by ensuring that only authorized users have access to sensitive data. This can help organizations avoid fines and other penalties.

Principle of Least Privilege

The principle of least privilege (PoLP) is a security concept that states that users should only be granted the permissions they need to perform their job functions. This helps to protect systems and data from unauthorized access and use.

Protect Data and Systems

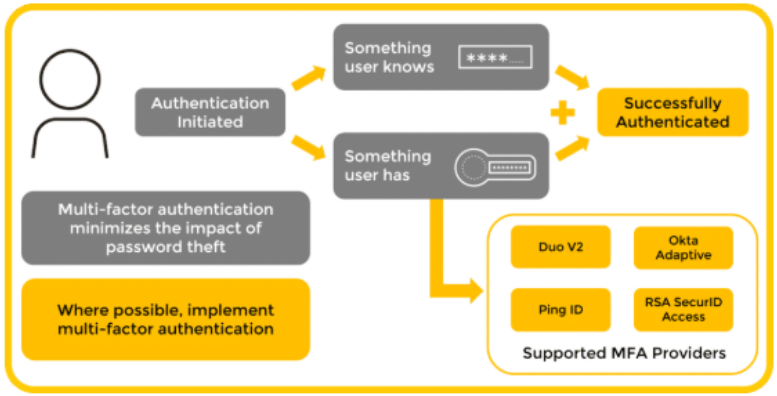
IAM aims to protect sensitive data and systems from unauthorized access.

Single Factor Authentication

The firewall can use external servers to control administrative access to the web interface and end user access to services or applications through GlobalProtect. In this context, any authentication service that is not local to the firewall is considered external, regardless of whether the service is internal (such as Kerberos) or external (such as a SAML identity provider) relative to your network.

Multi-Factor Authentication

When multi-factor authentication is enabled, a user or an attacker must present two or more forms of user credentials, called factors, to gain access to a network resource. The first factor commonly is a username and password. The additional factors often are some type of numerical code that is generated on a mobile phone app or on a dedicated security key fob, or by software installed on the user's laptop or desktop system.



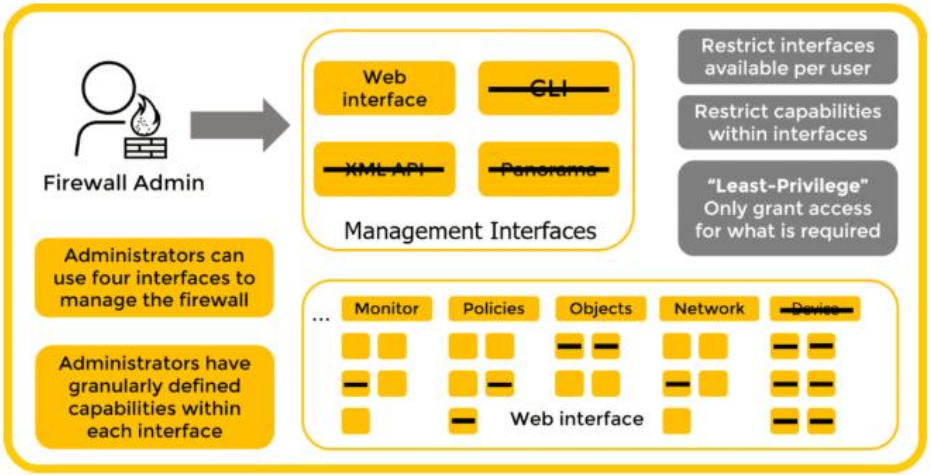
Authentication Policy

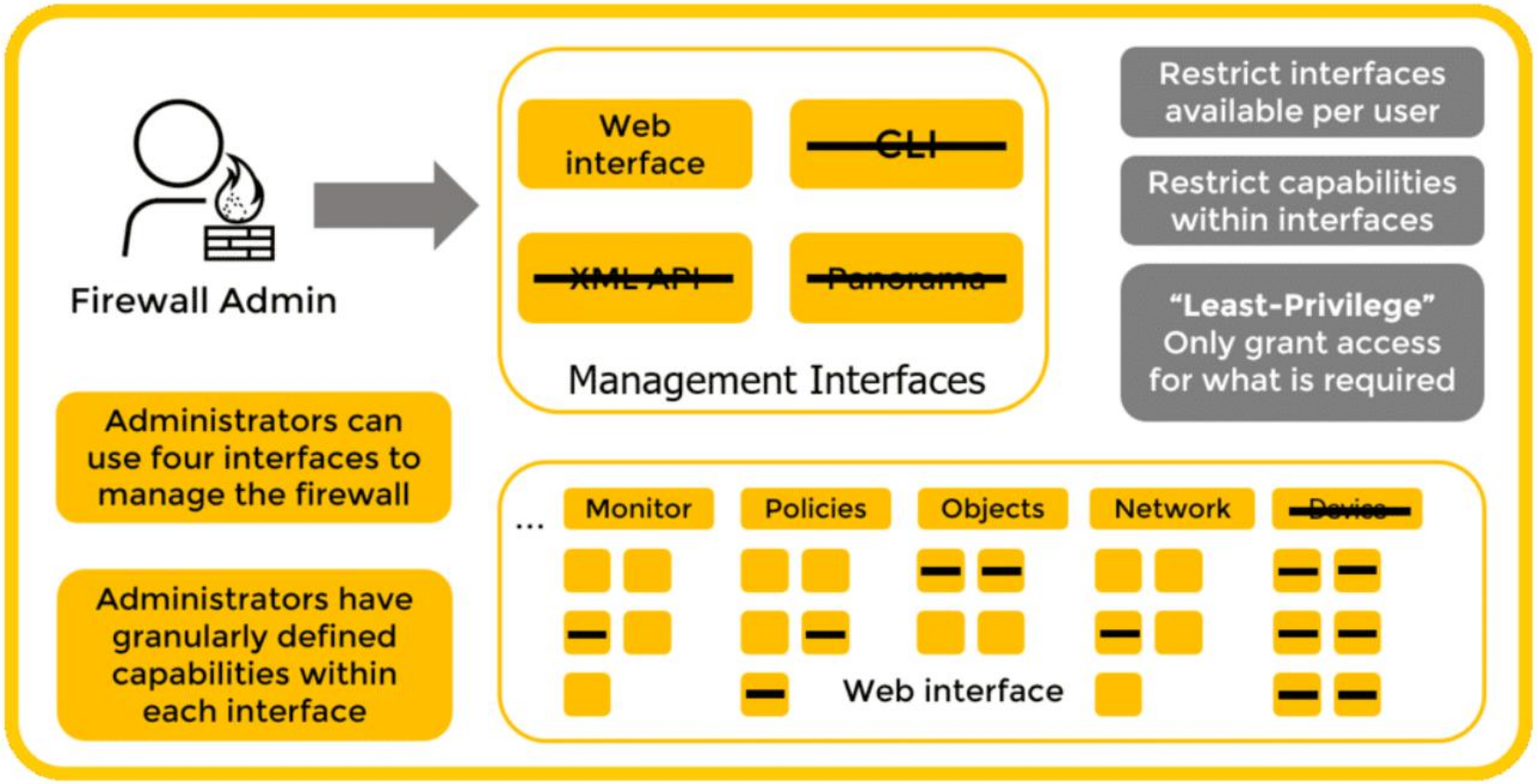
An Authentication policy enables an administrator to selectively issue multi-factor authentication challenges based on the sensitivity of the information stored on the network resource. A firewall administrator also can configure the number and strength of the factors of authentication based on the sensitivity of the information on each network resource. For example, you could require all corporate users to authenticate using multi-factor authentication once a day but require IT administrators to use multi-factor authentication each time they use Remote Desktop Protocol (RDP) to access an Active Directory server.

Role-Based Access Control

Role-based access control (RBAC) is an access control model that restricts access to resources based on the roles that users hold within an organization. In RBAC, users are assigned to roles, and roles are granted access to resources. This allows administrators to manage access to resources by simply managing the roles that users belong to.

When you create a custom role, use the "least privilege" approach to grant user access. Restrict interfaces available per user and capabilities within each interface of the web interface.




Firewall Admin
Web interface
CLI
XML API
Panorama
Management Interfaces
Restrict interfaces available per user
Restrict capabilities within interfaces
"Least-Privilege" Only grant access for what is required
Administrators can use four interfaces to manage the firewall
Administrators have granularly defined capabilities within each interface
...
Monitor
Policies
Objects
Network
Device
Web interface

Other Types of Access Control

1. Attribute-based access control (ABAC) provides and manages user access to IT services, supporting areas that require more contextual awareness than simple user-focused parameters like an assigned role.

2. Discretionary Access Control (DAC) gives the application owner complete control over who can access a particular service, such as a file, directory, or other network-accessible resource.

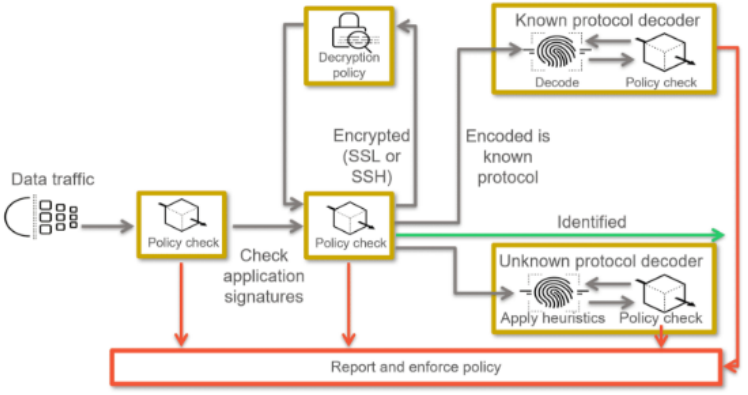
3. Mandatory Access Control (MAC) enforces access to resources through a security policy controlled by the operating system, offering higher security than DAC but with increased complexity in implementation and management.



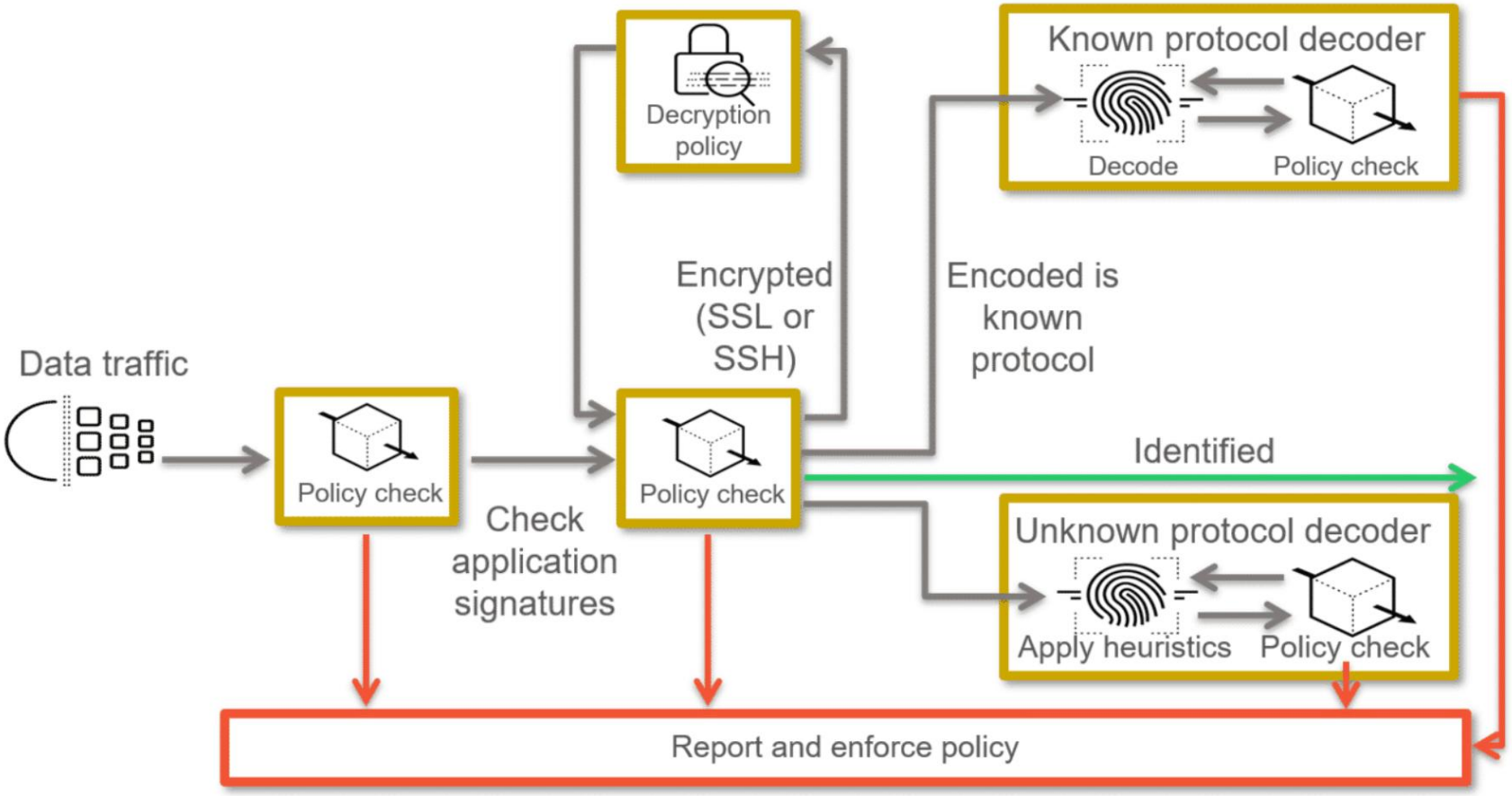
Port-based stateful packet inspection technology was created more than 25 years ago to control applications using ports and IP addresses. Using port-based stateful inspection to identify applications depends on an application strictly adhering to its assigned port(s). This presents a problem because applications can easily be configured to use any port. As a result, many of today's applications cannot be identified, much less controlled, by the port-based firewall, and no amount of "after the fact" traffic classification by firewall "helpers" can solve the problems associated with port-based application identification.

App-ID Architecture

Palo Alto Networks App-ID technology does not rely on a single element, such as a port or protocol. Instead, App-ID uses multiple mechanisms to determine what the application is. The application identity then becomes the basis for the firewall policy that is applied to the session. App-ID is highly extensible, and application detection mechanisms can be added or updated to keep pace with the ever-changing application landscape.



Click to enlarge.


Known protocol decoder
Decode
Policy check
Decryption policy
Encrypted (SSL or SSH)
Encoded is known protocol
Data traffic
Policy check
Policy check
Identified
Check application signatures
Unknown protocol decoder
Apply heuristics
Policy check
Report and enforce policy

User-ID

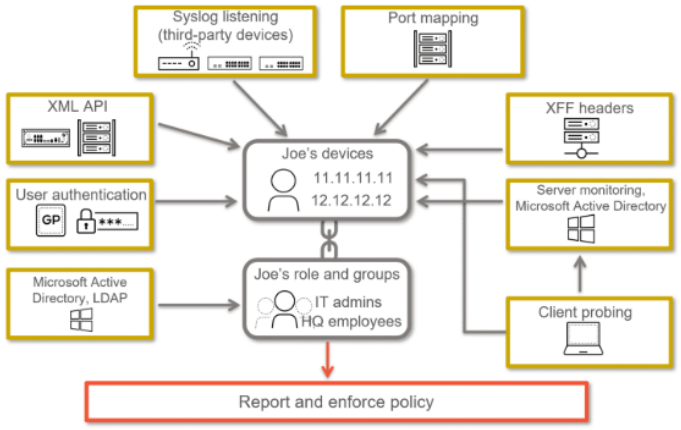
The next-generation firewall accurately identifies users for policy control.



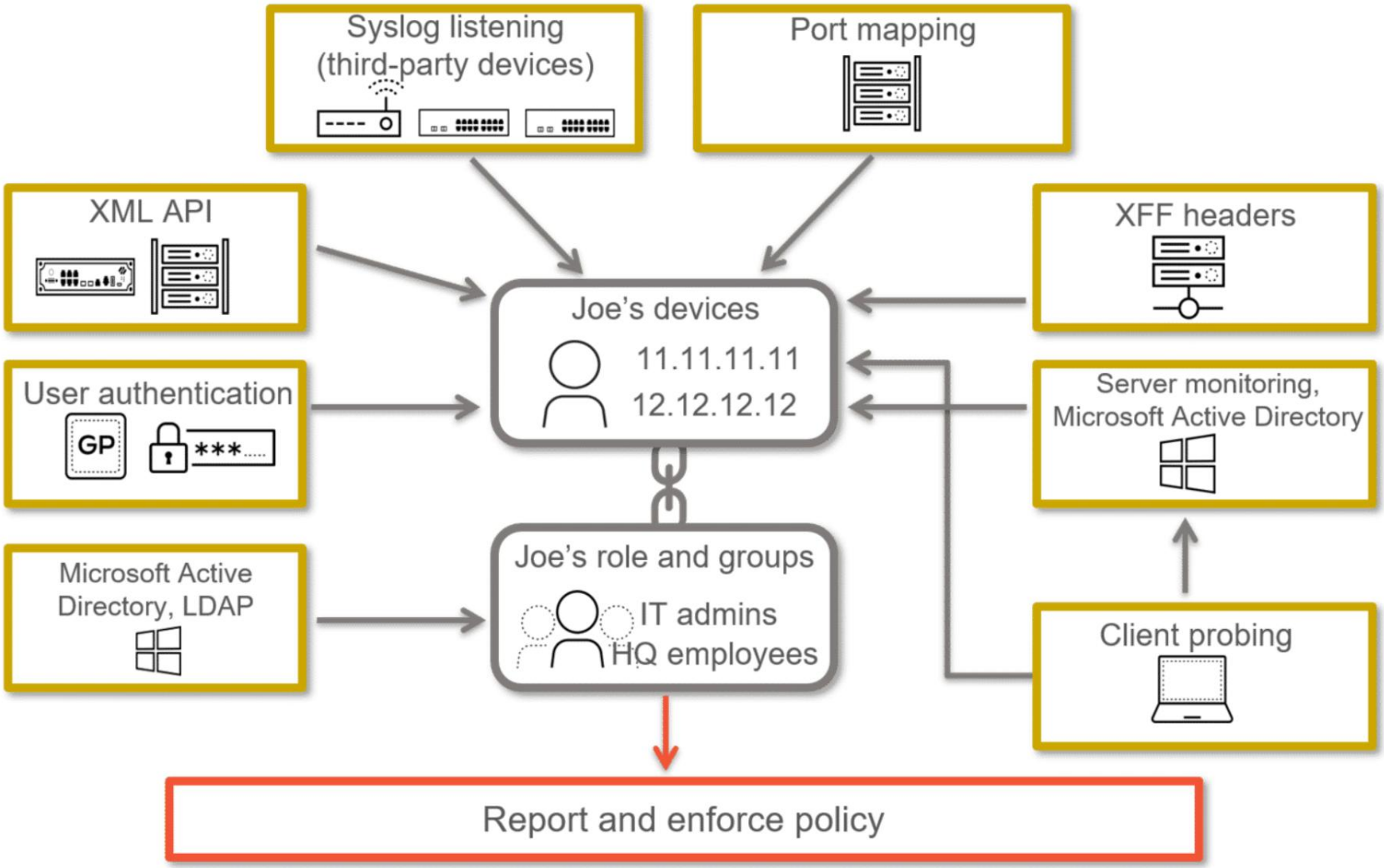
A key component of security policies based on application use is identifying the users who should be able to use those applications. IP addresses are ineffective identifiers of users or server roles within the network. With the User-ID and Dynamic Address Group (DAG) features, you can dynamically associate an IP address with a user or server role in the data center. You can then define user- and role-based security policies that adapt dynamically to changing environments.

User-ID Architecture

In environments that support multiple types of end users across a variety of locations and access technologies, it is unrealistic to guarantee physical segmentation of each type of user. Visibility into the application activity at a user level, not just at an IP address level, allows you to more effectively enable the applications traversing the network. You can define both inbound and outbound policies to safely enable applications based on users or groups of users.



Click to enlarge.


Syslog listening
(third-party devices)
Port mapping
XML API
XFF headers
Joe's devices
11.11.11.11
12.12.12.12
User authentication
GP
***
Server monitoring,
Microsoft Active Directory
Joe's role and groups
IT admins
HQ employees
Microsoft Active
Directory, LDAP
Client probing
Report and enforce policy

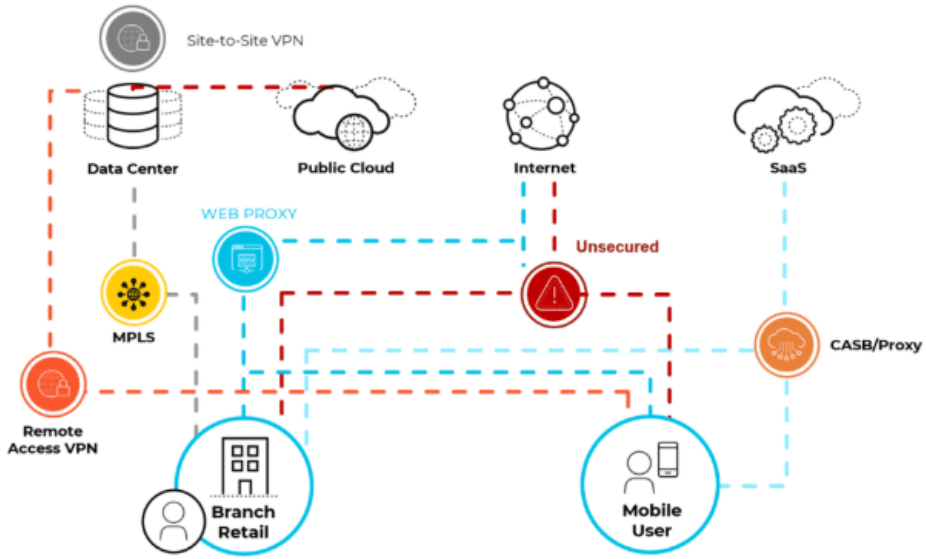
Secure Access Service Edge (SASE)

Introduction

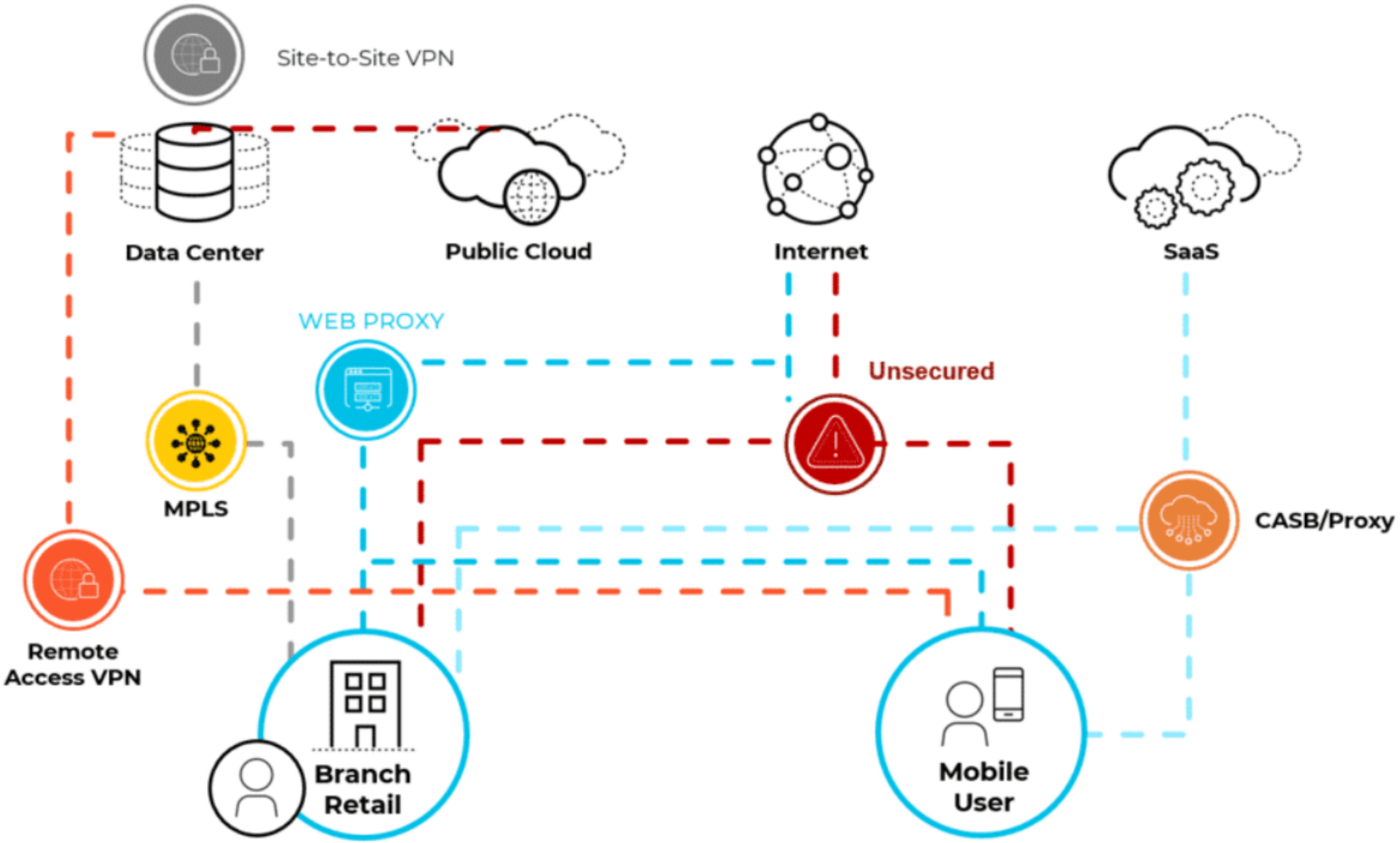
This module analyzes the benefits of an enterprise Secure Access Service Edge network implementation. A successful SASE solution secures the connections of users, devices, and applications to cloud-based resources and applications.

Today, many of the technologies on the market are built upon architectures that were not designed to handle all types of traffic and security threats. This forces organizations to adopt multiple-point products to handle different requirements, such as secure web gateways, firewalls, secure virtual private network (VPN) remote access, and software-defined wide area network (SD-WAN).

For every product, there is an architecture to deploy, a set of policies to configure, and an interface to manage, each with its own set of logs. This situation creates an administrative burden that introduces cost, complexity, and gaps in security posture.



Click the image to enlarge it.


Site-to-Site VPN
Data Center
Public Cloud
Internet
SaaS
WEB PROXY
Unsecured
MPLS
CASB/Proxy
Remote
Access VPN
Branch
Retail
Mobile
User

SASE - Complete Visibility Across Hybrid Environments

SASE enables complete visibility of hybrid enterprise network environments that connect data centers, headquarters, branch and retail locations, public and private cloud, and mobile users.



The combination of Firewall as a Service (FWaaS,), Secure Web Gateway -(SWG), Data Loss Prevention (DLP), and Cloud Access Security Broker (CASB) capabilities and functions in SASE provides enterprise security teams with full visibility into all network activity in the environment, including users, data and apps.

SASE - Granular Control of Users, Data, and Apps

Users are increasingly leveraging a variety of applications — including SaaS applications from multiple devices and locations — for work-related and personal purposes. SASE provides a refined security focus on all user, data and application activities.

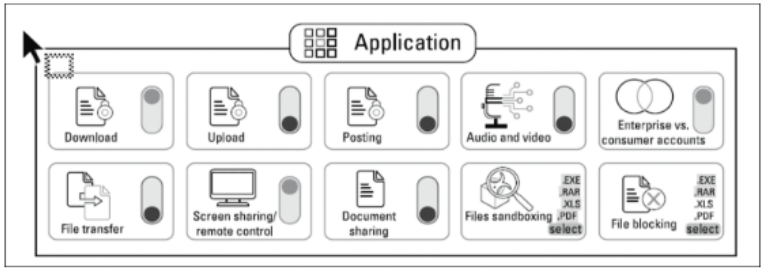
How Users Might Introduce Risks When Using Applications

Nonstandard Ports

Many applications, such as instant messaging (IM), peer-to-peer (P2P) file sharing, and Voice over Internet Protocol (VoIP), are capable of operating on nonstandard ports or hopping ports. Some of these applications are sanctioned by the organization, others are tolerated, and others are unsanctioned.

Users Violate Policies

Users are increasingly savvy enough to force applications to run over nonstandard ports through protocols such as Remote Desktop Protocol (RDP) and Secure Shell (SSH), regardless of the organization's policy regarding various applications (sanctioned, tolerated, unsanctioned).



This diagram shows a variety of applications users access. Click to enlarge.

SASE Remedy - Classify Traffic

SASE provides complete visibility into application usage. It provides capabilities to understand and control application use by configuring appropriate policies and rules. SASE can classify application traffic by ports, by default — thereby reducing the administrative burden of requiring research for which applications use which ports.

SASE - Reduced Complexity

SASE enables your business to simplify networking and security by:



Eliminating unnecessary, limited use of siloed point security solutions.



Operating from the cloud ato cut operational complexity and cost.



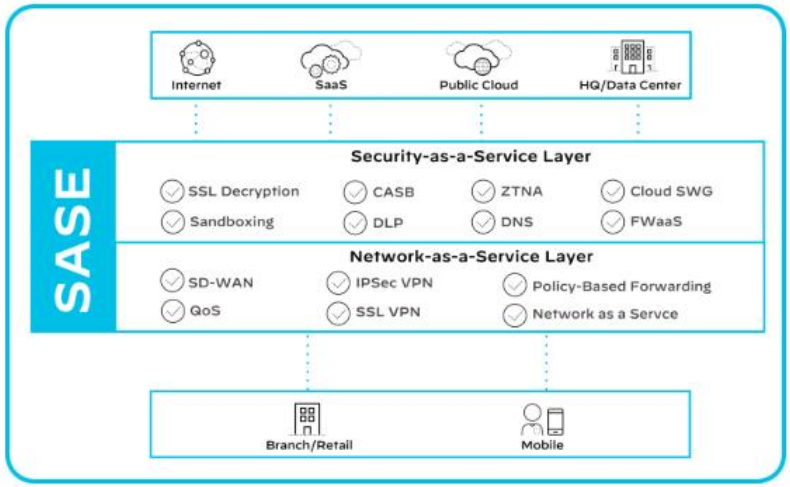
Avoiding logistical issues with shipping, installing, and upgrading multiple networking and security hardware devices to remote branch (or retail) locations.

Introducing Prisma SASE
Simplifying Cloud Security and Access

Prisma SASE is designed to help organizations embrace cloud and mobility by providing network and network security services from a common cloud-delivered architecture.

A SASE solution must provide consistent security services and access to all types of cloud applications (public cloud, private cloud, and software as a service (SaaS)) delivered through a common framework. By removing multiple-point products and adopting a single cloud-delivered Prisma SASE solution, organizations can reduce complexity while saving significant technical, human, and financial resources.

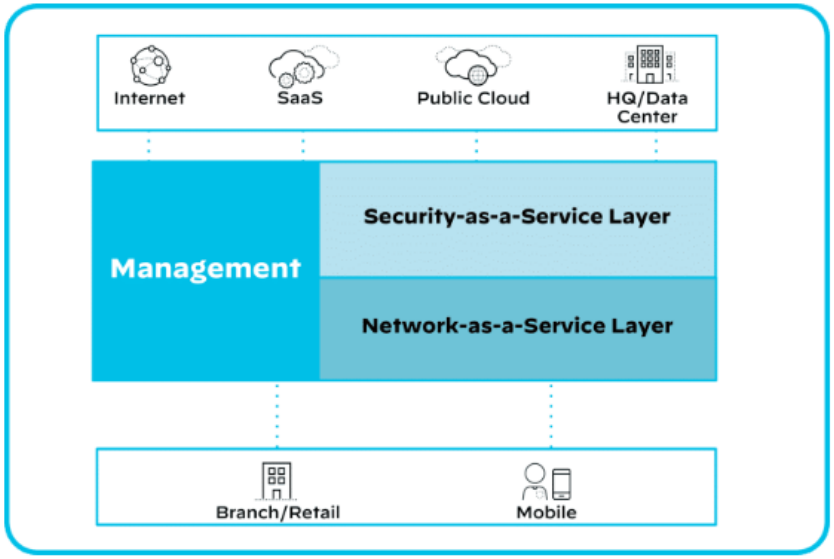
A SASE solution converges networking and security services into one unified, cloud-delivered solution that includes security as a service and network as a service layers.



Click the image to enlarge.

Components of Prisma Access SASE Architecture

The Prisma Access SASE architecture consists of a network-as-a-service layer, a security-as-a-service layer, and a common management platform to secure branch/retail and mobile users across SaaS, public cloud, internet, and headquarters/data center environments



Click the image to enlarge.